OECD ECONOMIC SURVEYS

LUXEMBOURG

1997

SPECIAL FEATURES

- LABOUR MARKET
- PRODUCT MARKET COMPETITION

OECD ECONOMIC SURVEYS

1996-1997

LUXEMBOURG

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

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This Survey is based on the Secretariat's study prepared for the annual review of Belgium and Luxembourg by the Economic and Development Review Committee on 27th and 28th January 1997.

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After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 21st February 1997.

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The previous Survey of Belgium and Luxembourg was issued in September 1995.

BASIC STATISTICS OF LUXEMBOURG (1995)

THE LAND

Area (sq. km)	2 586
Agricultural area, 1994 (sq. km)	1 267
Woodland, 1992 (sq. km)	886

Major city, thousand inhabitants:	
Luxembourg, 1.3.1991	75.8

THE PEOPLE

Population (thousands), 31.12.1995	412.8
Inhabitants per sq. km	160
Net natural increase	1 624
Net migration	4 610

Employment (thousands):	
Total domestic employment	213.1
Dependent employees:	197.1
Agriculture	1.6
Industry and construction	56.3
Services	139.2
Employers, self-employed persons and domestic help	16.0

PRODUCTION

Gross domestic product (billion LF)	513.5
Gross domestic product per head (US$)	42 172
Gross fixed investment:	
Per cent of GDP	24.2
Per head (US$)	10 200

Gross domestic product by origin, at market prices (per cent):	
Agriculture	1.0
Industry and energy	16.3
Construction	6.7
Other	76.0

THE GOVERNMENT

Per cent of GDP:	
General government consumption	12.2
Central government current revenue	30.2
General government gross debt	6.0

Composition of the Chamber (number of seats):	
Christian-socials	21
Socialists	17
Democrats	12
Others	10
Total	60

Last election: 12.6.1994

THE CURRENCY

Monetary unit: Luxembourg franc

Currency units per US$, average of daily figures:	
Year 1996	30.975
January 1997	33.072

Note: An international comparison of certain basic statistics is given in an annex table.

LUXEMBOURG

Assessment and recommendations

Firm growth supported by services continued in 1996

Industrial production weakened in 1996, but owing to a rapid expansion of the service sector, GDP is estimated to have risen by almost 4 per cent. The external sector was the major force behind the buoyancy of the economy. Although exports of goods declined, this was more than compensated by a sharp increase in exports of services, mainly originating in the financial sector. Moreover, import growth decelerated due to the weakness of the manufacturing sector, and thus the contribution of the foreign balance to growth increased. Consumption growth seems to have accelerated, despite almost stagnant national employment and weak real wage growth. However, investment lost most of its buoyancy, due to a sharp decline of fixed investment in the manufacturing sector (outside the steel industry) and the termination of a special project.

Employment remains robust, while wage inflation slows

Employment growth remained robust both in 1995 and 1996, with most jobs created in the service sector. As most vacancies were filled by cross-border workers, unemployment hardly changed. Non-employment continued to be relatively high partly due to the growing number of people in early retirement and disability schemes. Wage inflation decelerated, following a steady decline in consumer price inflation, a reduction in employers' social

security contributions, and increased competition from cross-border workers.

Sustained growth is expected

GDP growth is projected to remain close to 4 per cent in both 1997 and 1998. Exports are expected to accelerate, reflecting a stronger performance of the steel industry, and more buoyant economic conditions in neighbouring countries. The contribution of the foreign balance is thus likely to stay firmly positive. Investment could pick up in 1997 but consumption growth will continue to lag behind GDP growth. Employment growth is expected to remain strong, despite continued job losses in the steel sector. Consumer price inflation may edge up modestly to 2 per cent, partly due to higher import prices. However, activity in the steel sector and the economy in general could be less buoyant than projected, especially if the recovery of the European economies is weaker than expected.

Public sector finances remain healthy

Public finances were in surplus in the first half of the 1990s and Luxembourg easily satisfies the Maastricht criteria. Budget policy is set in a medium-term framework whereby most of central government expenditure is subjected to an expenditure growth norm. Nevertheless, public spending has been growing fast – notably due to a rapidly growing social security sector – which partly reflects the legal indexation of many expenditure items and the rising number of benefit claimants. However, owing to buoyant tax receipts, the public sector surplus could even increase to around 2 per cent in 1996. According to government projections, public sector finances are projected to deteriorate slightly in 1997, but will remain in surplus.

The Government has given priority to maintaining the attractiveness of Luxembourg as a site for industrial activities with a multi-annual investment programme aimed at improving the infrastructure and supporting the start-up of

new industries, with a growing emphasis on research and development (R&D). The problem of rising labour costs has been partly mitigated by reducing employers' social security contributions. Taxes, as a percentage of GDP, are among the highest in the OECD but this can be partly attributed to characteristics specific to Luxembourg, notably the presence of a large and profitable financial sector. The Government recently reduced corporate taxes to respond to tax breaks offered by other financial centres, thereby maintaining the country's competitive edge in the financial sector. Further reductions are planned for the coming years.

Reforms are needed to meet future budget claims

Although budgetary problems are not expected in the near future, additional initiatives may be useful to restrain expenditure growth. The rigidity of public spending could become a problem, even more so as spending demand, related to the ageing population, is expected to accelerate. In addition, it is uncertain how long the current buoyancy of corporate tax receipts will continue. Faced with these challenges, and considering the dependence of the economy on a few sectors, it would be prudent to keep government expenditure in check. Public finance management could be improved by enhancing the efficiency of the public sector, a stricter application of the expenditure norm, and a reduction in the widespread use of indexation which would improve the flexibility of the budget.

Labour market outcomes appear favourable but non-employment is rising

The Luxembourg labour market has been characterised by strong employment growth, low unemployment and a rising number of foreign workers. However, the official unemployment numbers do not include the large and rising number of people who receive benefits for which no job search is required – such as early retirement and disability benefits. The authorities have stepped up initiatives – notably aimed at reducing labour costs for specific groups,

increasing training positions and improving the functioning of the Public Employment Service. These measures do not seem to have had a significant impact on national employment levels, partly because of high unemployment in adjacent regions. However, rigidities in the labour market and the lack of work incentives also appear to have reduced their effectiveness.

Increase labour market flexibility

Although the current situation does not call for drastic action, some reforms may be useful to increase the flexibility of the labour market and improve labour market participation. Due to rigidities rooted in the wage formation system, wages have been insufficiently responsive to labour market conditions, notably at the lower end of the wage scale. To widen the wage structure and bring wages more in line with productivity, notably for low-skilled workers, wage indexation should be abolished and the role of minimum wages as a distributional instrument reassessed. Furthermore, relaxation of the administrative extension of sectoral wage agreements would allow outsiders to price themselves back into the market. Employment for low-skilled workers could also be improved by further reducing the already low taxes and social security contributions at the bottom of the wage scale. In addition, employment chances for outsiders could be improved by relaxing regulations concerning fixed-term contracts and Sunday working.

Shift budgetary means to employment promotion

The Luxembourg welfare system is among the most generous in the EU for resident workers but is likely to have reduced work effort and incentives to look for work. Replacement rates are high, notably at the minimum wage level, and benefit durations in some cases are unlimited. In general, the government could improve incentives by scaling back generosity of the social security system, thus free-

ing up budgetary means to support programmes that promote employment. Work incentives could be improved by reducing the withdrawal rate for additional earnings in the mean-tested general assistance scheme (RMG) or introducing in-work benefits. More importantly, the authorities should review the housing assistance which is currently conditional on receiving an RMG benefit. De-coupling both benefits could reduce the marginal effective tax rates for RMG claimants substantially.

Strengthen the link between job search and benefits

In Luxembourg, many people receive benefits for which no job search is required – such as disability and early retirement schemes. The government should tighten access to these programmes and strictly limit the disability schemes to people who are genuinely disabled and scale back the use of the early retirement schemes. Also the work requirement in the RMG should be more strictly enforced, notably for older claimants.

Summing-up

The Luxembourg authorities are operating in an enviable macroeconomic environment characterised by sustained economic growth, low unemployment, and a public finance surplus. In addition, inflationary pressures are absent, partly due to the role of the cross-border workers. However, a weak point is the strong dependence of the economy on a few sectors, notably financial services and the heavy reliance on foreign resources to generate income. Although current rigidities of the economy have not been an obstacle to robust growth, a reversal in the prospects – or even a slowdown – of the service sector could have profound effects on the economy. Thus structural reforms, notably in the labour market, may be required to improve the adaptability of the economy.

I. Recent trends and prospects

Demand and output

The Luxembourg economy grew by almost 4 per cent both in 1995 and 1996, and although lower than in previous years, the rate of growth was still well above the EU average (Table 1).[1] The good performance was the result of contrasting sectoral developments. Services, and notably a thriving financial sector, were the main driving force behind the economic expansion but manufacturing, and in particular the steel industry, was affected by the cyclical slowdown in the neighbouring countries. Despite robust growth over the last decade, the economy does not show signs of overheating. An important factor is the presence of a large pool of labour in the surrounding regions of Germany, France and Belgium, allowing the Luxembourg economy to rely on an ever growing number of cross-border workers.

Growth continued to be strong in service sectors, notably banking and insurance, communications and business services. The banking sector has remained buoyant, despite growing competition from other financial centres in Europe. As margin pressures have increased, many banking institutions have undertaken or announced a restructuring of their activities and shifted their focus to fee-generating business. Growth in financial services was especially pronounced in the insurance and re-insurance sector as the rapid incorporation of the European directives on insurance into Luxembourg law enticed many international groups to set up business in the country. Output in the communication sector was bolstered by the deployment of two new satellites for the provision of digital television services.

However, production in the manufacturing sector fell in the second half of 1995 and most of 1996, largely reflecting the weak conjunctural situation in continental Europe. The largest decline was in the steel sector, due to heavy

Table 1. **Demand and output: recent trends and projections**

Annual percentage change, 1990 prices

	1993 current prices		1995	1996	1997	1998
	LF billion	Per cent of GDP				
A. Demand and output						
Private consumption	257.0	58.4	2.4	2.9	2.8	2.5
Government consumption	56.9	12.9	2.2	4.1	3.9	1.5
Gross fixed investment	125.1	28.4	3.5	0.8	4.0	5.0
Final domestic demand	439.0	99.7	2.7	2.4	3.3	3.1
Stockbuilding [1]	–19.4	–4.4	0.4	0.2	0.2	0.2
Total domestic demand	419.6	95.3	3.1	2.7	3.5	3.3
Exports of goods and services	401.2	91.1	4.4	4.2	4.7	4.9
Imports of goods and services	380.6	86.5	3.8	2.8	4.2	4.3
Foreign balance [1]	20.6	4.7	0.8	1.5	0.9	1.0
GDP at constant prices	. .	. .	3.7	3.9	4.1	4.0
GDP price deflator	. .	. .	1.0	2.2	2.1	2.2
GDP at current prices	440.1	100.0	4.7	6.2	6.3	6.2
B. Memorandum items						
Private consumption deflator	. .	. .	0.7	1.5	1.8	2.0
Industrial production	. .	. .	1.8	–2.0	2.2	2.5
Total employment	. .	. .	2.5	2.6	2.6	2.5
Unemployment rate (per cent)	. .	. .	3.0	3.3	3.3	3.2

1. Contribution to growth of GDP.
Source: OECD Secretariat.

destocking by major foreign customers and increased competition from eastern European countries. The contraction in steel production may have bottomed out in the second half of 1996, owing to an end to destocking. Production in other manufacturing sectors was also affected by the cyclical downturn, but the contraction was limited as a result of the strong performance of certain industries (*e.g.* wood processing). Production in the construction sector (on a value added basis) progressed steadily by around 2½ per cent both in 1995 and 1996. However, national construction sector output has been on a declining trend since 1993, reflecting growing competition by non-resident firms whose share in total sales rose from 5 per cent in 1985 to 15 per cent in 1995.

On the demand side, exports seem to have been the major force behind the current buoyancy of the economy. After gathering momentum in the first half

of 1995, exports of goods decelerated sharply during the rest of that year and in 1996, largely reflecting the evolution of steel shipments. However, this was more than compensated by a sharp increase in exports of services (11 per cent in 1996), notably fee-generating banking services. On the import side, lower imports of intermediate goods by the steel industry were partly offset by higher imports of investment goods. Overall, the contribution of net exports to GDP growth may have widened significantly in 1996, to some 1½ percentage points.

Investment developments can be rather volatile in a small economy like Luxembourg, where specific investment projects can make a significant impact on overall investment. In 1995, investment was boosted by several projects that are less sensitive to the business cycle, like the construction of electric furnaces in the steel industry, the launch of a communication satellite, and the purchase of two aircraft. Although the effect of some of these factors continued in 1996, business investment was almost stagnant, following the termination of an air-cargo centre in the previous year. Moreover, investment in the manufacturing sector (outside the steel industry) sharply dropped. Residential investment seems to have strengthened in 1996, after being sluggish in the previous year. Government investment slowed but remained at a relatively high level, supported by the multi-annual investment programme.

As in previous years, growth in private consumption – at 2.4 per cent – was well below GDP growth in 1995, reflecting almost stagnant national employment and slowing real wage growth. For 1996, consumption growth is estimated to have accelerated to almost 3 per cent, well beyond the real growth of wages, resulting in a fall in the savings rate. Car purchases increased sharply in the first ten months of 1996 although part of these purchases were re-exported. Public consumption accelerated to around 4 per cent in 1996.

Employment, wage and price inflation

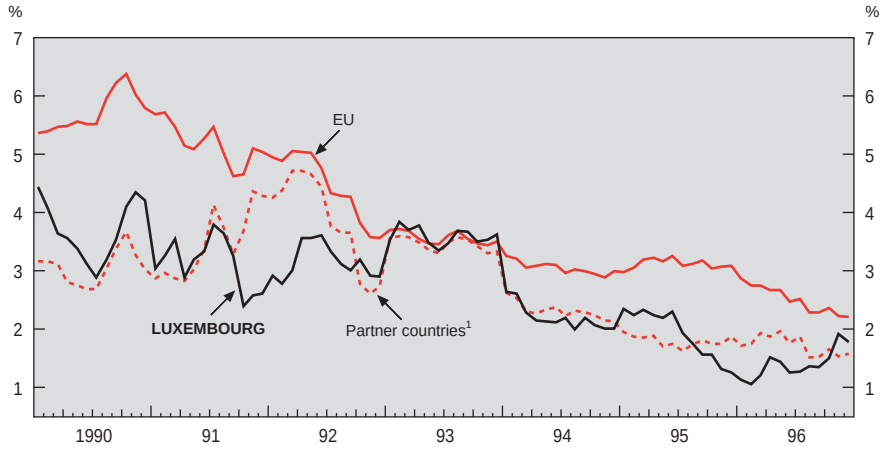
Employment growth has remained robust, amounting to around 2½ per cent both in 1995 and 1996. As most vacancies were filled by cross-border workers, their share in total employment increased to over 25 per cent; in the financial services, hotel and catering, and construction sectors, their employment share is even approaching 40 per cent. The increase in employment was sharpest in the service sector, although job creation in the banking sector – partly as a result of

outsourcing – decelerated. Due to the reorganisation of the steel industry, employment in this sector is still on a downward trend. On average, 700 jobs were lost in both 1995 and 1996, amounting to almost 10 per cent of the workforce in this industry in each year. Unemployment – at around 3 per cent – remains low compared to other OECD member countries, although non-employment stays at a relatively high level. Labour market developments and policies are discussed in greater detail in Chapter III.

Consumer price inflation has been on a downward trend since 1991 and reached 1.4 per cent in 1996, which was somewhat lower than in the surrounding countries (Figure 1). The low rate of inflation largely reflects slow price increases for imported goods, which make up 60 per cent of the consumer price index (CPI) basket. Prices for food, clothing and footwear hardly changed, partly due to sluggish retail sales. On the other hand, housing costs were almost 6 per cent higher than a year earlier. The falling rate of inflation contributed, through the institutionalised wage indexation (*échelle mobile*), to a slower pace of wage increases.

Figure 1. **CONSUMER PRICES**

Per cent change over previous year



1. Belgium, France and Germany.

Source: OECD, *Main Economic Indicators.*

Table 2. **Compensation per employee**

Percentage changes

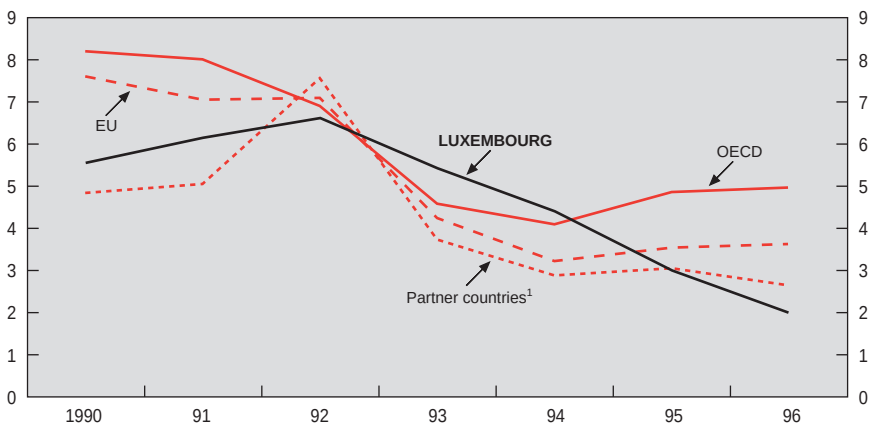
	1990	1991	1992	1993	1994	1995	1996
Compensation per employee	5.6	6.1	6.6	5.4	4.4	3.0	2.0 [1]
of which:							
Mining and manufacturing	2.7	6.1	4.1	4.7	4.5	3.2	. .
Construction	5.2	8.6	6.8	4.3	3.1	3.0	. .
Market services	1.3	10.5	6.0	5.6	3.2	4.1	. .
Banking and insurance	5.8	7.1	9.7	8.9	11.6	4.8	. .
Non-market services	13.3	2.8	8.7	3.7	3.6	1.2	. .
Employer's contribution	0.0	0.0	0.1	0.0	–0.8	–0.8	0.0
Gross wage	5.6	6.1	6.5	5.4	5.2	3.8	2.0 [1]
Wage indexation (*échelle mobile*)	3.7	3.1	3.2	3.6	2.2	1.9	1.0
Real wage (including wage drift)	1.8	2.9	3.2	1.8	3.0	1.9	1.0
Memorandum item:							
Minimum wage	3.3	9.6	5.0	7.1	3.5	2.1-5.2 [2]	0.8

1. Estimates.
2. The distinction between workers with and without dependent family members was abolished in 1995.

Source: Statec, Inspection générale de la Sécurité sociale and OECD Secretariat.

Figure 2. **COMPENSATION PER EMPLOYEE: AN INTERNATIONAL COMPARISON**

Growth rate, per cent



1. Belgium, France and Germany.

Source: Statec and OECD Secretariat.

Compensation of employees has been decelerating since the early 1990s, but was higher for most of the first half of the 1990s than in neighbouring countries (Table 2 and Figure 2).[2] In 1995, the slowdown was partly due to the carry-over of the abolition of the employer's contribution to the child benefit scheme (*allocation familiale*) in July 1994. In 1996, the major effect came from a deceleration in the *échelle mobile*, whose contribution was almost 1 percentage point lower than in the previous year. Also increased competition from cross-border workers had a mitigating effect on wage settlements. Wages in the financial sector showed the fastest increases in the first half of the 1990s, partly due to wage drift, and notably increases in bonuses. In the steel industry, average compensation was affected by the replacement of low-skilled jobs with high-skilled jobs due to the introduction of electric furnace steelmaking.

The balance of payments

The value of exports of merchandise goods weakened in the second half of 1995 and early 1996 due to the downturn in activity in continental Europe and falling prices. In the steel industry – which accounts for about one-third of goods exports – sales of long products were down by 13 per cent in the first half of 1996 due almost entirely to weaker prices. Exports of plastics and rubber products increased by 15 per cent in 1995 but have stagnated since the beginning of 1996. At the same time, imports of intermediate goods for the steel industry also contracted. However, the decline of imports was limited by a surge in imports of transport equipment in 1995 and 1996 related to the delivery of several aircraft. Overall the trade balance deficit rose considerably, to almost 13 per cent of GDP in the first three quarters of 1996.

The visible trade deficit was more than offset by substantial surpluses on the balances of factor incomes and other services (Table 3). The service balance improved in 1995, partly due to an increase in air transport of passengers (6 per cent) and cargo (18.5 per cent). The banking sector's contribution to the rise of the service balance was due to the increase in fee-earning activities rather than earnings on interest margins. On the other hand, rising numbers of cross-border workers led to a further increase in income and transfers paid to non-residents. Overall, the current account surplus in 1995 was 18 per cent of GDP.

Table 3. **Current balance of payments**

LF billion

	Average 1990-92	1993	1994	1995[1]
Current balance of payments	100.3	79.1	90.8	92.8
As a per cent of GDP	27.1	18.0	18.9	18.1
Goods and services				
Goods	–53.2	–63.0	–58.7	–50.9
Exports	211.8	201.2	216.6	267.5
Imports	265.0	264.2	275.3	318.4
Services	37.7	45.6	46.1	78.7
Exports	90.4	111.5	121.7	283.2
Imports	52.7	65.9	75.6	204.5
Income (net)	123.9	105.4	115.4	80.0
Compensation of employees	–19.4	–31.4	–37.5	–44.0
Investment income	143.3	136.8	152.9	124.0
Current transfers (net)	–8.1	–8.9	–12.0	–15.0
General government	. .	. .	. .	–4.1
Other sectors	. .	. .	. .	–10.9
Memorandum item:				
Current balance including reinvested earnings	57.2	60.8	68.1	. .

1. Provisional data, break in series.
Source: Statec.

Short-term prospects

Real GDP growth is projected to remain close to 4 per cent in both 1997 and 1998. Exports are expected to be buoyant, reflecting *inter alia* the economic upturn in neighbouring countries which should end the massive destocking by major steel customers. As a result, the contribution of the foreign balance to growth is likely to stay firmly positive. On the domestic side, gross fixed investment could become the most dynamic component of total demand again. In 1997, the progressive switch from the integrated steelmaking process to the more efficient electric steelmaking process will be completed when the last blast furnace is replaced by an electric one. Business investment will also be boosted by the expansion of the Astra satellite network with four satellites under construction and further launches planned in 1997 and 1998. However, due to their large import content, the direct effect of these investments on GDP growth will be only

marginal. Government investment is projected to remain strong due to school building and investment in infrastructure – notably, rail infrastructure and an acceleration in road building beginning in 1998.

The growth of private consumption is projected to lag behind GDP growth, in line with the evolution of real wages and a slight increase in national employment. Government consumption is likely to grow rapidly in 1997 as Luxembourg assumes the Presidency of the EU in the second half of the year. Employment growth is expected to remain robust, but as most new jobs are likely to be filled by cross-border workers, the unemployment rate may decline only slightly. However, employment in the steel sector is set to be reduced from almost 7 000 in 1996 to 4 000 in 2000, mainly by early retirement. Wages are expected to remain strong, especially in the financial sector and the public sector following the multi-annual wage agreement. Higher import prices might push up consumer price inflation to 2 per cent. The main risk to the projections relate to the external side: a weaker than expected recovery in continental Europe would significantly depress economic activity in Luxembourg, especially in the steel industry.

II. Economic policies

Monetary policy

Owing to the Belgium-Luxembourg Economic Union (BLEU), Luxembourg does not have an independent monetary policy. Monetary policy for the BLEU is conducted by the National Bank of Belgium (NBB).[3] Since mid-1995, the NBB has repeatedly reduced its interest rates, generally in line with the Bundesbank. Hence money market rates have remained very close to German rates, falling by some 150 basis points from June 1995 to January 1997. The decline in long-term rates has been of the same order, as the interest differential with Germany has been virtually eliminated. The OECD Secretariat does not expect a major change in monetary conditions over the projection period. Short-term rates may start rising later in 1997, in line with German rates and reflecting an acceleration in real GDP growth. However, long-term rates may be somewhat lower in 1997-98, on average, than in 1996.

As discussed in some detail in the previous *Survey*,[4] the legislation governing the Luxembourg Monetary Institute (LMI) is being modified to meet the requirements of the Treaty on European Union, *i.e.* the so-called Maastricht treaty, to transform the LMI from a virtual monetary authority into a fully fledged central bank of Luxembourg, operating in the framework of the European Union and as a member of the European system of Central Banks.

Budgetary policy

Luxembourg's fiscal performance compares favourably with that of other OECD Member countries. In the first half of the 1990s, public sector finances were in surplus, with central and local government finances close to balance, while the social security sector showed a substantial surplus due to the building

Table 4. **General government budget and debt**

SNA basis, LF billion

	1992	1993	1994[1]	1995[1]	1996[2]
General government net lending	3.1	7.4	12.6	7.8	10.6
As a per cent of GDP	0.8	1.7	2.6	1.5	2.0
Central government	–5.5	0.1	4.0	–1.4	0.7
Local government	–2.1	–0.8	1.1	1.1	0.0
Social security	10.6	8.1	7.5	8.1	9.9
General government gross debt	20.5	27.1	27.6	30.8	35.3
As a per cent of GDP	5.2	6.2	5.7	6.0	6.6
Memorandum items:					
Reserves of special state funds[3]	8.5	11.4	15.0	18.7	. .
State budgetary reserves[4]	6.1	2.7	2.8	3.3	. .

1. Provisional.
2. Estimates.
3. *Avoir des fonds de l'État.*
4. *Réserves budgétaires de l'État.*

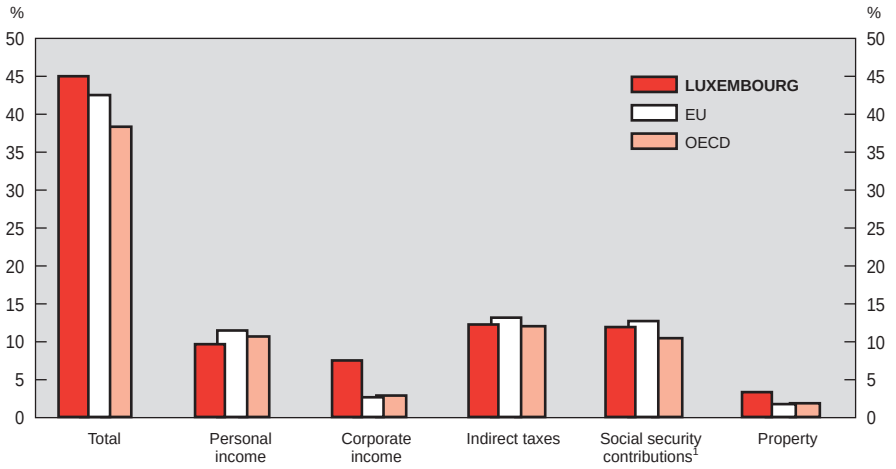
Source: Inspection générale des finances.

up of the reserves of the pension scheme (Table 4). The consolidated public debt (Maastricht definition), although slightly increasing, was no more than 6 per cent of GDP during most of the early 1990s. However, public sector spending has been growing fast, notably due to a rapidly expanding social security sector where spending rose by more than 10 per cent per annum in the early 1990s. At the same time, total tax revenues (including social security contributions) – as a percentage of GDP – increased from 43.6 per cent in 1990 to 45 per cent in 1994, and are among the highest in the OECD (Figure 3).[5]

The high tax burden is partly attributable to characteristics specific to Luxembourg, *e.g.* a substantial part of value added tax (VAT) and excise duties is paid by non-residents who – attracted by lower prices – do their shopping in Luxembourg. In addition, corporate tax revenues – at 7.6 per cent of GDP – are substantially higher than the OECD average. This is only partly due to the level of tax rates which are somewhat higher than in the EU on average. However, the rules determining taxable income are rather flexible, in particular for the financial sector, owing to the generous tax treatment for provisions and technical reserves, and tax receipts are boosted by the large and profitable banking and insurance sectors.

Figure 3. **TAXES AND SOCIAL SECURITY CONTRIBUTIONS**

1994, as a per cent of GDP



1. Including payroll tax.

Source: OECD (1996), *Revenue Statistics 1965-1995.*

The central government's budgetary policy is set in a medium-term framework, aimed at balancing government finances, and linking expenditure to GDP growth. The growth of expenditure is determined by a ''growth norm'', reflecting the expected medium-term real growth rate of the economy and expected movements of the *échelle mobile*. Overall spending by the special funds[6] and by the social security sector is not included in the expenditure norm, but government transfers to the special funds are. In practice, the norm has been applied in a flexible way as increases in transfers to the social security sector and some exceptional expenditure were not subjected to the budget norm. The government can also temporarily increase spending above the budget norm by running down the assets of the special funds, which are not subjected to the budget norm. However, owing to the favourable development of tax revenues, budget outcomes have usually been better than initially projected. As actual GDP growth frequently exceeded the medium-term real growth rate in the first half of the 1990s, central government expenditure (excluding special funds) came down by 3 percentage points, to around 29 per cent of GDP in 1995.

The Government has given priority to maintaining the attractiveness of Luxembourg as a site for financial and industrial activities. This is partly done through a regularly updated multi-annual investment programme, mainly aimed at improving the transportation infrastructure, environmental protection and education. Government investment – at more than 4 per cent of GDP for the period 1994-96 – is already higher than in any other EU country. The government's industrial diversification policy aims at slowing down the de-industrialisation process by supporting the start-ups of new industries, notably in the steel producing areas, which may partly compensate employment losses in the steel works. Under the current industrial diversification law, the emphasis has shifted from direct investment support to measures favouring R&D spending. In order to improve international competitiveness and stimulate employment, employers' contributions were reduced, and are among the lowest in the OECD. To make up for income losses, the Government has steadily increased its share in the funding of the social security sector. Moreover, corporate tax rates were reduced in early 1997 in response to tax breaks offered by other financial centres, thereby maintaining the country's advantage in the financial sector. Further reductions are planned for the coming years.

Recent developments

The public sector financial surplus was 1.5 per cent of GDP in 1995, compared to a projected surplus of 0.6 per cent. Strong profits in the banking sector, where lower provisions against doubtful loans to developing countries resulted in higher tax liabilities, gave rise to larger than projected company tax receipts by almost LF 4 billion (0.8 per cent of GDP). In addition, VAT revenues were higher than expected by LF 1 billion.

The 1996 central government budget projected an increase in primary expenditure of 8 per cent, compared to an expenditure growth norm of 5.1 per cent (Table 5). The overrunning of the norm was caused by expenditure increases which were not subjected to the budget norm, such as a transfer to the child benefit scheme (*allocations familiales*) of almost LF 1 billion – to cover a chronic deficit in the scheme. Also the rapid increase in investment spending – in line with the multi-annual investment programme – was for a substantial part not subjected to the budget norm as it was financed by reducing financial assets of the special investment funds. The budget did not include significant changes in

Table 5. **Central government budget**[1]

LF billion

	1994 Final	1995 Provisional	1996 Budget	1997 Budget
Primary receipts	143.3	149.6	159.0	. .
of which:				
Indirect taxes	62.0	64.6	70.6	. .
Direct taxes	75.0	77.7	80.4	. .
Primary expenditure	142.4	154.8	167.2	. .
of which:				
Consumption	41.3	48.0	50.9	. .
Transfers to other sectors	29.0	23.7	27.2	. .
Transfers to local government and social security	50.6	60.2	63.4	. .
Investment and capital transfers	21.4	22.9	25.7	. .
Primary surplus	0.9	–5.1	–8.2	. .
Net interest payments	–3.8	–3.9	–3.9	. .
Financial balance	4.7	–1.2	–4.3	. .
Memorandum items:				
Budget growth norm	5.2	5.2	5.1	3.9
Expenditure growth (excluding special funds)	6.4	6.3	5.7	4.3

1. National definition. This table is not comparable with Table 4 which is on an SNA basis.
Source: Inspection générale des finances.

tax policies and primary receipts were projected to grow by 6 per cent, below the nominal GDP projection for that year. Although the draft budget deficit implied a widening of the central government deficit, a small surplus for the overall public sector was projected.

Budget outcomes are likely to be better than projected, especially on the revenue side. Although the economy has decelerated and growth is likely to be around 1 percentage point below the official growth projection, income taxes and indirect taxes – the latter due to an acceleration in VAT receipts – were substantially higher in the first half of 1996 than a year earlier. In addition, a favourable execution of the 1995 EU budget resulted in the reimbursement of LF 1.1 billion in EU contributions. Overall, receipts could be more than 1.5 per cent of GDP higher than projected, thus increasing the estimated public sector surplus to 2 per cent of GDP.

A new wage agreement for the public sector has been concluded for the period 1995-99. Under the new agreement, real wages will be increased by around 1 per cent per annum, including a gradual increase in the end-of-year bonus. The new wage agreement also includes revisions to the pension scheme. Civil servants will be completely compensated for increases in their pension contributions, which will be steadily increased from 3 per cent in 1994 to 8 per cent in 1999. It will bring pension contributions on a par with those in the private sector. The changes will only have an accounting impact, since pensioners in the public sector receive their pensions directly from the government.

The 1997 Budget and beyond

According to the official projections public sector finances will remain in surplus in 1997. Details for the 1997 budget of the central government (including the special funds) are still not fully available. The 1997 expenditure norm has been set at 3.9 per cent. This does not include LF 0.7 billion for exceptional expenses due to Luxembourg's EU Presidency in the second half of 1997. Employment growth in the government and semi-government sectors is projected to decelerate, thereby mitigating the impact of the multi-annual wage agreement. Wages and salaries – adjusted for changes in the financing of the nursery and primary education system[7] – are expected to rise by about 3.5 per cent. Overall, current spending is projected to increase by 5.5 per cent, well above the budget norm, thereby squeezing capital spending in the central government budget (excluding the special funds). However, investment spending will again be kept in line with GDP growth by reducing assets of the special investment funds. The Government announced company tax cuts – equivalent to LF 4.1 billion in 1997 – to encourage foreign investment in Luxembourg and preserve the competitiveness of the economy. The tax cuts include the abolition of the local working capital tax (*Gewerbekapitalsteuer*) and the reduction of the corporate income tax by 1 percentage point a year from 33 per cent in 1996 to 30 per cent in 1999. The Government will only partly compensate the local authorities for the loss in the working capital tax.

The current fiscal position of Luxembourg is healthy and budgetary problems are not expected in the near future. With public sector finances in surplus and gross public sector debt at around 6 per cent, the country easily meets the criteria for participation in EMU. A potential problem is the rapid expansion

of government expenditure which, in the past, was met by fast growth in tax revenues, for a substantial part related to the large and profitable financial sector. However, it is uncertain how long the current buoyancy of corporate tax receipts will continue. Less favourable outcomes for the financial sector could have serious repercussions for the budget. The government might have to offer more tax breaks to the financial sector to head off the growing competition between financial centres. In addition, further tax harmonisation in the EU might also diminish the competitive advantage of Luxembourg in financial services.

The authorities have found it difficult to slow down expenditure growth due to the rigidity of the budget: about two-thirds of expenditure are – through the *échelle mobile* – directly linked to changes in the consumer price index while many other budgetary items are also adjusted for price increases. Moreover, the multi-annual wage agreement for public servants has added to the rigidity of the budget by fixing real wage increases until 1999. As in most other European countries, the authorities are faced with an ageing population which will progressively raise the demand for social spending. Faced with these challenges and considering the dependence of the economy on a few sectors, it would be prudent to keep government spending in check. The authorities should aim at improving the efficiency of the public sector, for instance, by introducing performance indicators, and should reassess its role notably in industrial policy. In addition the medium-term budget norm could be more strictly applied and its application widened to include also the social security sector and special funds. Reducing the widespread use of indexation would also increase the flexibility of the budget.

III. Implementing the OECD Jobs Strategy

Introduction

Over the past two decades or so, the Luxembourg labour market has been characterised by strong employment growth and low unemployment. These favourable labour market outcomes have been mainly the result of the relatively high growth rates of the economy. One of the paradoxes is that despite strong employment growth, unemployment in various forms (*i.e.* ''broad unemployment'') has increased since the early 1990s as most of the new jobs have been filled by foreign workers – and recently mainly cross-border workers. The apparent mismatch between the supply of domestic jobs and the demand from the resident population,[8] points to the possible emergence of structural problems in the labour market. The most important of these seem to be: the rigid wage formation system, which leads to a very compressed wage distribution with high labour costs at the lower end of the market; the lack of incentives to work as a result of the generous social welfare and tax systems; and strict employment protection rules. Another important feature has been the large proportion of people who have withdrawn from the labour market by entering early retirement and especially the disability schemes. As a result, the participation rate of older workers is one of the lowest among Member countries, and the country's employment rate is below the OECD average. The Government has stepped up initiatives – largely targeted at the lower end of the labour market – to improve the unemployment situation. While there is no need for drastic action, it may be prudent to use the current period of economic strength to take further measures to make the labour market more flexible.

This chapter follows up the general structural policy recommendations of the *OECD Jobs Study* with specific recommendations for Luxembourg. The first section presents an overview of the Luxembourg labour market and employment

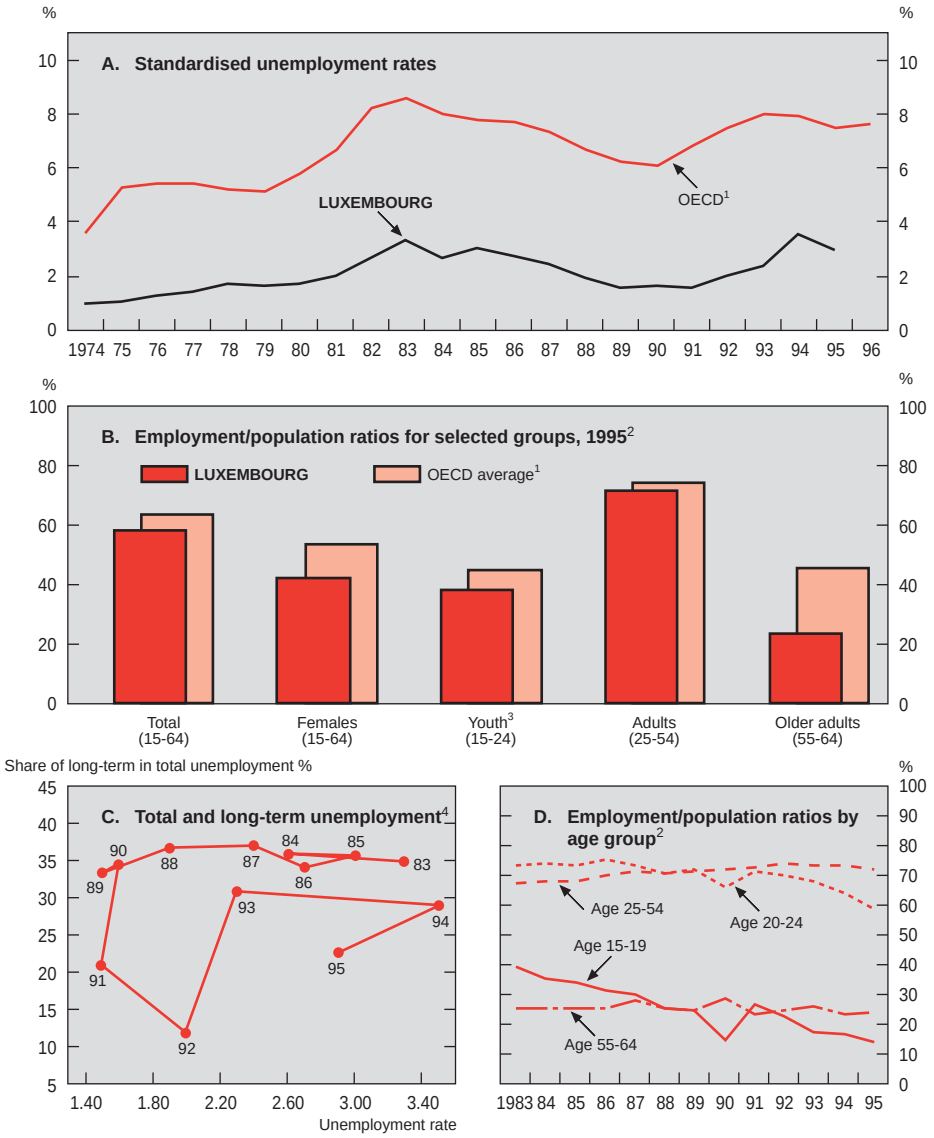
performance over the past two decades. The policy requirements for Luxembourg and their motivations are discussed in the second section, followed by a review of recent policy actions and an assessment of the scope for further action.

Labour market and employment performance

Many aspects of the labour market performance compare favourably with other OECD countries (Figure 4, Panel A), with unemployment at only 2.9 per cent (according to a standardised definition)[9] in 1995. This has been only partly due to the impressive employment growth, with domestic employment recording an average annual growth of 3¾ per cent since the early 1970s (Figure 5, Panels A and B). In addition, many people left the labour force, due to the large-scale restructuring of the steel industry in the second half of the 1970s and in the 1980s. Employment in the steel industry fell over this period from almost 24 000 to around 7 000, which was achieved virtually without adding to unemployment (Figure 6). Older workers made use of a government-funded early retirement programme while other supernumerary workers found work in public work schemes (*division anti-crise)*.

The Government has pursued an active industrial policy by stimulating new economic activities. The initial aim of this diversification policy was to reduce the economy's dependence on the steel industry and facilitate the establishment of new manufacturing industries, but it has gradually widened to include other sectors, notably services. Over the last two decades, the economy underwent a remarkable transformation from an industrial economy – dominated by the steel industry – to a service economy, and Luxembourg has progressively become an important financial centre. It has done this by exploiting its comparative advantages in the form of a strict secrecy code, tax advantages over neighbouring countries and liberal financial rules and regulations combined with a favourable geographical location at the heart of Europe and a qualified and multilingual labour force. Later, the insurance sector – spurred by the creation of a single European market in insurance – and the communications sector have also become major growth areas. The share of the service sector in total employment increased from 35 per cent in 1970 to over 65 per cent in the 1990s (Figure 5, Panel C).

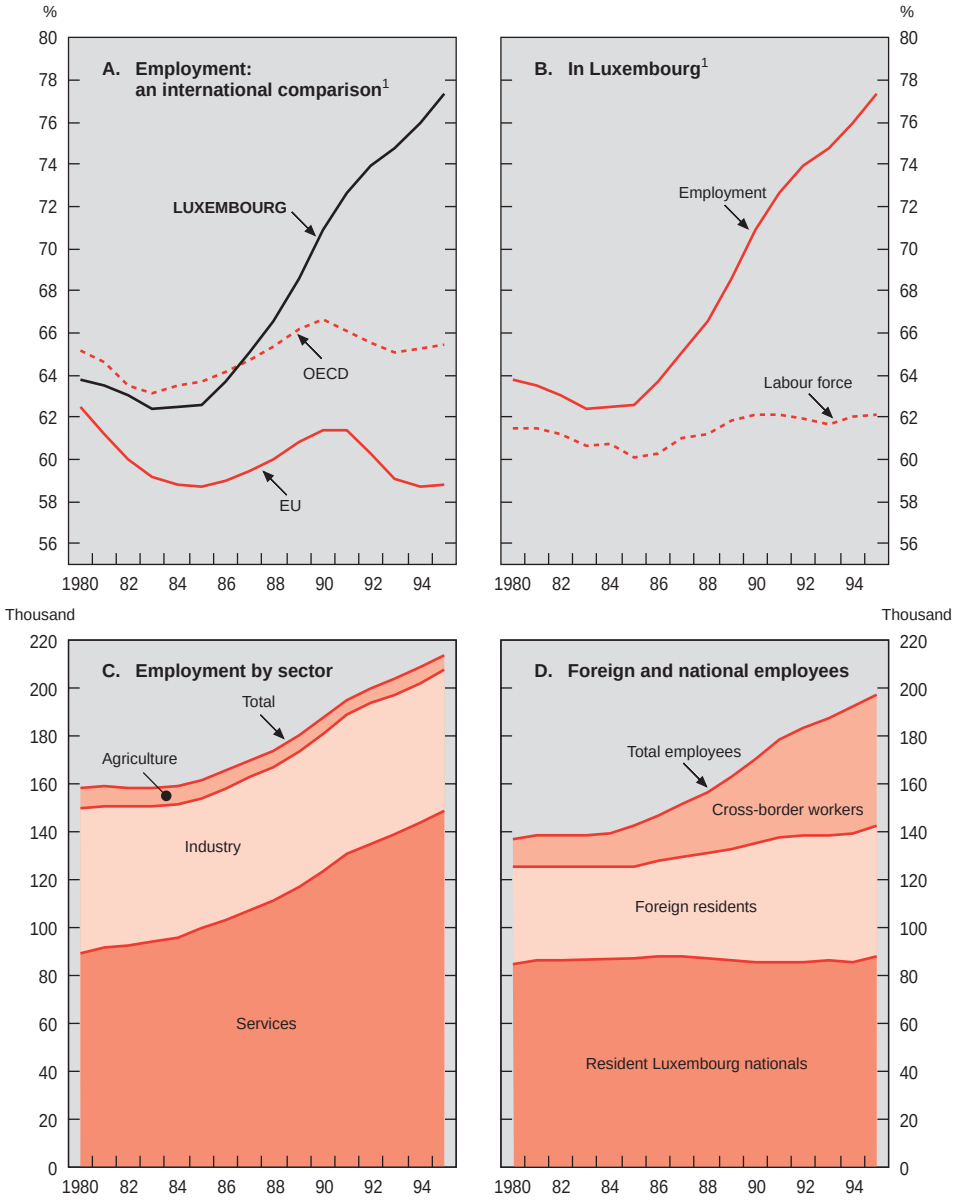
Figure 4. **EMPLOYMENT AND UNEMPLOYMENT**



1. The number of countries included in the OECD average varies depending on the data available.
2. Defined as the percentage of each population group that is employed.
3. The minimum age for youth differs across countries (either 14, 15 or 16), therefore, any comparison with the OECD average should be made with care.
4. Long-term unemployment is defined as individuals looking for work for one year or more.

Source: OECD, *Labour Force Statistics, Main Economic Indicators* and OECD Secretariat.

Figure 5. **OVERVIEW OF THE LABOUR MARKET**

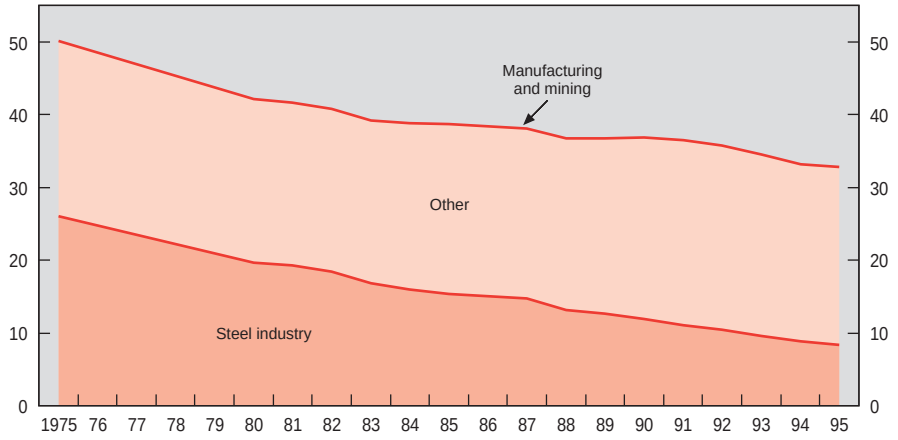


1. As a per cent of the working age population.

Source: Statec, *Statistical Yearbooks*; Inspection générale de la Sécurité sociale; and OECD Secretariat.

Figure 6. **EMPLOYMENT IN THE STEEL INDUSTRY**

Thousand persons



Source: Statec.

The shift in activities to the service sector has affected the composition of the labour force. Due to the restructuring of the steel industry, many – mostly male – workers took early retirement. On the other hand, the substantial increase in jobs in the service industry created more job opportunities for women and their share in domestic employment rose to around 35 per cent in 1995, from 25 per cent in 1975. Nevertheless, the female participation rate of the resident population – at 36 per cent – remains low compared with other OECD countries. In part, this may reflect low incentives to work due to high marginal effective tax rates for some groups (see below). Many women have taken part-time jobs, and between 1992 and 1995 female part-time employment rose by 3 percentage points to about 20 per cent. On the other hand, male part-time employment remained stable at around 1 per cent over the same period. Hence, overall part-time employment still represents only 8 per cent of the national labour force.[10]

The increase in unemployment since the early 1990s has been largely due to a mismatch between supply and demand for labour, against a backdrop of relatively inflexible wages. Around 60 per cent of those seeking employment

have a low level of education (less than a completed secondary education) while demand for unskilled labour has been substantially reduced because of technological change, notably in the steel industry and the banking sector. Unemployment duration has decreased compared to the 1980s (Figure 4, Panel C) and the share of long-term unemployment – at only 23 per cent in 1995 – is one of the lowest in the OECD.[11] The low incidence of long-term unemployment can be partly attributed to active labour market programmes. People unemployed for more than one year and falling back on the general assistance scheme agree with the benefit agency to participate in training or a subsidised work programme. Almost a third of older unemployed (older than 50 years) are registered with the Public Employment Service (PES) for more than a year. Health problems, insufficient skills and a lack of geographical mobility are often the cause of long-term unemployment for this group. As in many other European countries, the youth unemployment rate has been relatively high – almost twice the overall unemployment rate. However, partly due to various measures, their unemployment spells are relatively short and only 5 per cent are unemployed for more than a year. In addition, young people tend to enrol in education for longer periods of time, which may contribute eventually to the increase in the growth potential of the economy.

Traditional measures of unemployment fail to give a proper description of the Luxembourg labour market. The official unemployment numbers do not include people in active labour programmes, nor persons of working age who receive benefits that do not require any job search – such as early retirement[12] and invalidity benefits (Table 6). Disability benefits have probably become a substitute for unemployment benefits, notably in the steel industry. Disability benefits are attractive for employees – especially for those with no access to the early retirement schemes – as benefits are higher than unemployment benefits and unlimited in duration. Employers also profit from the scheme since it allows them to reduce their workforce easily without incurring expensive redundancy payments. The early retirement and disability schemes currently are much more important – in terms of persons enrolled – than the official unemployment scheme, and have removed a substantial part of the working-age population from the labour force. This is one reason why the employment rate is slightly below the OECD average and the participation rate for people older than 57 years old has fallen markedly over the past two decades and is now one of the lowest

Table 6. **Unemployment, special programmes and broad unemployment**

Annual averages

		1980-85	1990	1993	1994	1995
I.	**Unemployment**	2 077	2 060	3 526	4 643	5 130
	As a per cent of the national labour force	1.4	1.3	2.1	2.7	3.0
II.	**Other unemployed**					
	Part-time unemployed	126	42	64 [1]	23 [1]	31 [1]
III.	**Special labour market programmes**	2 412	627	1 245	1 339	1 082
	Job creation in steel sector (DAC)	2 216	101	650	600 [1]	171
	Youth work scheme (DAT)	114	380	432	467	558
	Training	82	146	163	272	353
IV.	**Other benefit programmes**	14 221	18 858	19 717	20 582	21 410
	Early retirement [2]	1 362	2 378	1 459	1 427	1 421
	Disability [3]	12 859	16 480	18 258	19 155	19 989
V.	**Broad unemployment** (I + II + III + IV)	18 836	21 587	24 552	26 587	27 653
	As a per cent of the broad labour force	11.1	11.8	13.0	13.9	14.3

1. Provisional or estimated data.
2. *Préretraite ajustement* or for shift workers.
3. End of year.

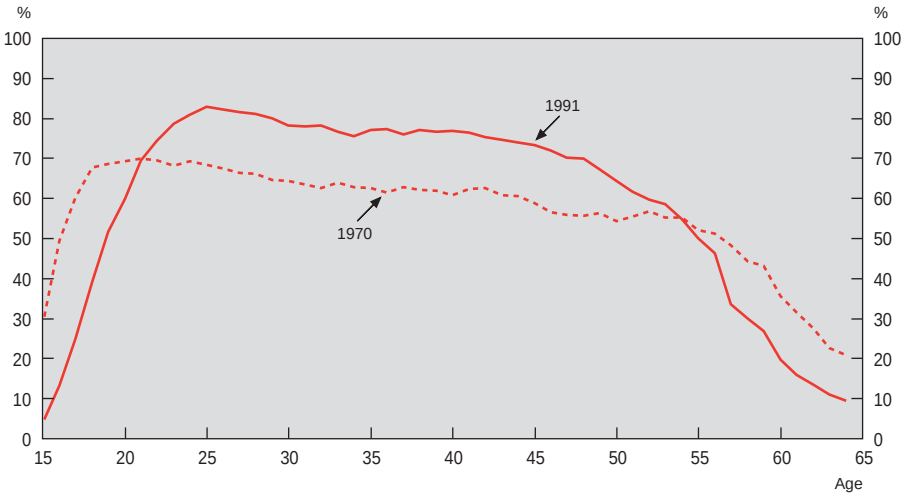
Source: Statec and Inspection générale de la Sécurité sociale.

among Member countries (Figure 4, Panel B and Figure 7). Hence, the concept of ‘‘broad unemployment’’ which includes unemployed, inactive persons of working age receiving a social security benefit and persons enrolled in special job creation programmes – while itself not entirely satisfactory and representing only a rough approximation – may provide useful additional information on the low participation rate. On this broad basis, unemployment has been steadily increasing, and was more than 14 per cent of the broad labour force in 1995.

A striking feature of the Luxembourg labour market is the importance of foreign workers, without whom the rapid expansion of the economy would not have been possible. The share of foreign nationals (excluding cross-border workers) in the population rose from almost 20 per cent in 1970 to 33 per cent in 1996, higher than in any other OECD country. The Portuguese are the single largest group, representing 10 per cent of the population, while nationals of adjacent countries represent 8 per cent. Participation rates of foreign nationals are relatively high: especially for youngsters and people older than 55 years for

Figure 7. **PARTICIPATION RATES**

Per cent



Source: Statec (1994), *Recensement de la population au 1er mars 1991*, Luxembourg.

whom rates are more than 10 percentage points higher than for Luxembourg nationals. Their share in domestic employment has risen to 25 per cent. A more recent trend is the growing incidence of cross-border workers in the Luxembourg economy, attracted by relatively high net salaries and faced with high unemployment in their own regions (Table 7). Their share in domestic employment has rapidly increased from less than 5 per cent in 1980 to 30 per cent in 1996. Hence, foreign workers – foreign residents and cross-border workers together – make up 55 per cent of domestic employment (Figure 5, Panel D).

The immigrant population is relatively young and more than 50 per cent were aged under 30 years old at the time of arrival.[13] Most immigrants who arrived during the period 1981-90, are working in the sectors of construction, hotel and catering, financial and other services. Relatively few of them have found work in the steel industry, mainly due to its restructuring. Also the transport and communication sector has received relatively few immigrants as most employment in this sector was reserved for Luxembourg nationals during this

Table 7. **Unemployment in adjacent regions**[1]

1993

	Germany	Belgium	France	Luxembourg
Active population (thousands)	478.7	137.4	328.6	167.5
Unemployment (thousands)	55.6	13.1	41.6	3.5
Unemployment rate (per cent)	11.6	9.5	12.7	2.1
Cross-border workers to Luxembourg (per cent of active population)	1.6	10.0	6.7	. .

1. Within 45 kilometres of the Luxembourg border.

Source: Fehlen F. and E. Jacquemart (1995), ''La main-d'œuvre frontalière au Luxembourg'', *Cahier économique* 84, Statec, Luxembourg.

period. Most cross-border workers come from France and are relatively young. The share of women has increased (from 20 per cent in 1975 to more than 30 per cent in 1993) following the expansion of the service industry. The education level of French cross-border workers is relatively high with 35 per cent of the female cross-border workers having completed at least an upper secondary education. Most male cross-border workers find employment in the manufacturing and construction sectors and have vocational qualifications.[14]

Policy requirements

In Luxembourg, in contrast to most other OECD countries, the rise in (broad) unemployment has coincided with strong domestic employment growth. Since the middle of the 1980s, in every single year the annual growth in the number of jobs would have been sufficient to absorb the total number of registered unemployed. However, most of the newly created jobs have been taken up by foreign workers, and more specifically by cross-border workers. It therefore seems that a strategy aimed simply at increasing employment will not have a significant impact on the rate of unemployment. The apparent mismatch between vacancies and demand by the resident labour force points to possible problems of qualifications, costs, incentives and more generally, problems in the functioning of the labour market which are explored below. The corrective measures discussed below and summarised in the Box are often complementary and mutually

Box. **The OECD Jobs Strategy: Synopsis of recommendations**

The *OECD Jobs Study* sets out a strategy based on nine recommendations for improving the ability of OECD economies to cope with structural change through enhancing the ability to adjust and to adapt, and increasing the capacity to innovate and be creative. The nine distinct policy areas covered included the macroeconomic policy framework and the creation and diffusion of technological know-how. With respect to labour and product market flexibility, it identified the need for initiatives in the following areas: working-time flexibility, the entrepreneurial climate, wage and labour cost flexibility, employment security provisions, active labour market policies, labour force skills and competencies, and unemployment and related benefit systems.

The Luxembourg labour market performance has been remarkably good, and in many respects the situation is quite different from that of other OECD countries. Nonetheless, the employment rate is low, and there has been a modest trend increase in unemployment and in ''broad unemployment''. The Luxembourg authorities have stepped up corrective initiatives, with the emphasis on the reform of the Public Employment Service, education reforms – including a smoother transition from school to work – and increasing the efficiency of existing programmes targeted at specific groups of unemployed.

Following from a review of the labour market and employment performance and the assessment of policy requirements, this chapter identifies the need for Luxembourg to move ahead in the following areas:

Increase wage and labour cost flexibility

- *Reform the wage bargaining system, phase out wage indexation and consider relaxing the administrative extension of sectoral wage agreements*. The *échelle mobile* leads to wage rigidity and hinders the proper functioning of the labour market. The administrative extension of sectoral wage agreements can be an obstacle for outsiders to price themselves back into the market: hence, as a minimum, opening clauses and wage scales close to the minimum wage should be introduced.
- *Reduce the minimum wage and its indexation* to widen the wage structure and allow wage rates to better match productivity levels, thereby improving employment prospects for low-skilled workers.
- *If for political or social reasons wage flexibility cannot be increased sufficiently, further reduce taxes and employers' social security contributions for low-skilled workers* since existing programmes only reduce non-wage labour costs for specific groups, which may lead to substitution.

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(continued)

Reform unemployment and related benefit systems

- *Reduce the generosity of the social security system and especially the unemployment benefit scheme, the early retirement schemes and disability scheme* to improve incentives for job search and, at the same time, free-up budgetary means for employment promoting programmes.
- *Reduce the withdrawal rate of benefits for additional earnings in the general assistance scheme (RMG) or consider the introduction of in-work benefits.* The present withdrawal rate provides an important disincentive for RMG recipients and their dependant relatives to look for work in the labour market and encourages black-market activities and long-term benefit dependency.
- *De-couple the housing assistance from the RMG.* The housing assistance scheme is the main reason that marginal tax rates can exceed 100 per cent and creates a major obstacle to the re-integration of RMG recipients in the active labour force.
- *Tighten access to early retirement and disability schemes.* Early retirement and disability schemes have been used as an alternative for the unemployment scheme. Access to the disability scheme should be strictly limited to people who are genuinely disabled, and the use of the early retirement schemes should be scaled back.

Expand and enhance active labour market policies

- *Apply more strictly the working requirement in the RMG scheme*, notably for older workers for whom there are special programmes to promote re-integration.

Increase employment flexibility

- *Relax employment protection legislation and ease restrictions on fixed-term contracts*. Although providing protection to the incumbent work force, current employment regulations may impair the employment prospect of outsiders. In particular, liberalising fixed-term contracts may enhance the prospects of the long-term unemployed finding a job.

reinforcing. For example, training programmes will be impaired if wage rigidity leads to a destruction of jobs at the lower end of the labour market or in the absence of incentives to accept such jobs. As unemployment remains low in Luxembourg, the present situation does not call for drastic action. Nevertheless, changes may be required to correct structural rigidities, increase the employment rate, and prevent future problems, such as those related to population ageing.

Enhancing the ability to adjust and adapt

The wage formation system

The significance of labour costs as one of the proximate causes for the emerging problems in the labour market has become more evident in recent years. Although employment creation has remained buoyant, employment prospects for low-skilled workers have worsened as many low-skilled jobs have been lost – mainly because of technological progress, notably in the steel industry and the banking sector. The wage formation system seems, at least until recently, to have been insufficiently responsive to worsening labour market conditions for this group. During recent tripartite consultations, the social partners have agreed to pay more attention to the competitive position of the firms in concluding collective wage agreements.

The rigidity of the wage formation process is rooted in the institutional and legal framework of wage setting, *i.e.* the administrative extension of collective wage bargaining agreements, full wage indexation[15] – the so-called *échelle mobile* – and the legal minimum wage. The minimum wage is also automatically adjusted in line with the *échelle mobile*. In addition, the minimum wage is reviewed at least every two years, in the light of the general economic situation and the evolution of average wages. Currently, the minimum wage is around 50 per cent of the average production worker's (APW) earnings.[16] According to the latest figures, 10 per cent of male employees and one-fifth of female workers receive the minimum wage.

The growing integration of labour markets in the *Grande région transfrontalière (GRT)*[17] has increased competition between resident and cross-border workers, exerting downward pressure on wages. Net wages in Luxembourg tend to be higher than in the surrounding countries, largely as a result of a substantially lower tax wedge. The tax wedge for the APW in Luxembourg is around 15 percentage points lower than in Belgium.[18] This makes low-skilled jobs in Luxembourg attractive for skilled workers from neighbouring countries. The tripartite Economic and Social Council (*Conseil économique et social*) recently expressed the fear that cross-border workers might replace resident workers.[19] There is no clear evidence that cross-border workers are paid less than residents for similar jobs. Although average earnings of cross-border workers are 18 per

cent lower than for resident workers this may be due to factors other than residency.[20]

While labour market legislation is extensive (*i.e.* indexation, minimum wages and job protection rules) the Government does not directly interfere in the wage bargaining process as such, and leaves it to the social partners. It considers a free wage bargaining process, embedded in the existing tight legal framework, a guarantee for social peace and its recent actions have been limited to reducing the tax wedge. From July 1994, employers' contributions to the child benefit scheme (*allocations familiales*) have been abolished and replaced by a government transfer to the scheme, reducing – *ceteris paribus* – total compensation by 1.7 per cent. In addition, tax wedges for specific groups in the local labour force have been reduced as part of active labour market programmes.

Taxes and transfers

The generosity of the social security system for Luxembourg residents and the structure of the tax system may have encouraged inactivity by reducing financial incentives to look for work. Luxembourg's social security system is among the most generous in the EU. Unemployment benefits are paid for a maximum of 365 days in a 24-month period and are means tested.[21] According to calculations by the OECD Directorate for Education, Employment, Labour and Social Affairs, the net replacement rate[22] in the first month of unemployment for an APW is 91 per cent for single family earners with two children, after Denmark the highest in OECD countries. Because of the limited duration, a relatively small percentage of the unemployed receive this insurance based benefit.[23]

The unemployment benefit is followed by a means-tested general assistance scheme (*Revenu minimum garanti*, RMG) for an unlimited period. The net replacement rate for an APW (including housing benefits) is 76 per cent. RMG claimants must be at least 30 years old and have been officially resident in Luxembourg for at least ten years out of the previous 20, a condition difficult to fulfil by foreign residents.[24] Partly because of that restriction, only 19 per cent of recipients are foreign nationals, which is significantly lower than their share in the overall population. Entitlement to RMG is in principle conditional on being available for work, but the vast majority of RMG recipients are exempted from this requirement because of age, incapacity or caring responsibilities. In some cases people over 50 years old are not required to be available for work if the

Public Employment Service has not offered them any suitable employment for three years.[25]

The level of RMG depends on the composition of the household and other income sources. An RMG benefit for a single person is 71.2 per cent of the legal minimum wage. Incomes from other household members are also taken into account in the determination of the benefit level, and can lead to substantial reductions. Above a relatively low threshold, the benefit withdrawal rate for additional earnings is 100 per cent.[26] Thus the scheme not only reduces incentives for the RMG recipient to look for work, but also discourages family members from any contact with the labour market. The disincentive problem is further strengthened by additional benefits like housing assistance and free public transport, which are linked to the RMG scheme. One-income families at the legal minimum wage can receive a means-tested RMG plus housing assistance, adding in some cases more than 60 per cent to their incomes. The marginal effective tax rate for this group is close to or even above 100 per cent[27] over a relatively wide range of earnings. This is a ''poverty trap'', as income is low but incremental earnings by any family member might leave them worse off. The system may lead to long-term benefit dependency and encourage black market activities. The lack of incentives, caused by the high marginal tax rates and the relatively generous benefit levels, may be one reason why integration programmes (see below) have not been successful as yet.

Increasing the flexibility of employment and working time

Luxembourg has some of the strictest employment protection rules in the OECD. Especially stringent are dismissal procedures and rules concerning fixed-term contracts and probation periods. As stressed in the *OECD Jobs Study*,[28] the effects of these rules on employment levels can go in two directions. On the one hand they may reduce hiring costs by setting general rules and standards; encourage firm-specific training and human capital formation; and provide for early notification of job loss, thus allowing employees to search for other employment prior to dismissal. In Luxembourg, the negative impact of firing on employers' costs are mitigated by the part-time unemployment scheme.[29] The clear advantage of the scheme is that workers are not dismissed and remain in contact with the labour market. Also, as noted, early retirement and disability schemes have been used as instruments to avoid expensive dismissals.

On the other hand, the current legislation may have been detrimental to the employment chances for disadvantaged workers (often low-skilled, long-term unemployed and youth). A cross-country analysis among OECD countries indicates that strict employment protection legislation has contributed to the persistence of unemployment.[30] In Luxembourg, restrictions on fixed term contracts[31] might have an adverse effect on the hiring of long-term unemployed. Surveys in other countries show that employers are reluctant to employ long-term unemployed, because of their lack of work experience and high dismissal costs if they prove unsatisfactory.[32] Moreover, strict employment protection rules have affected the structure of employment as firms are reluctant to hire permanent staff and make increased use of temporary workers, thus avoiding the high costs of terminating labour contracts.

Sunday work is strictly regulated and can only be introduced after reaching agreement with the existing workforce and obtaining permission from the Employment Minister. This could be an obstacle to outsiders wishing to enter the labour market. Recently the Employment Minister allowed Sunday working in a rubber processing plant – despite the opposition of the majority of workers – to secure an important investment programme. However, legislation concerning Sunday working in the retail sector was further tightened from February 1997.[33]

Active labour market policies

The generous social security system is combined with a wide palette of active labour market programmes, some of them targeted at specific groups – notably young people, long-term unemployed, women returning to the labour market, and disabled persons. These programmes have been regularly updated following developments in the labour market. Unemployment benefits can be made – at least in theory – conditional on participation in one of these programmes. In general, participants receive a compensation linked to their unemployment benefit or the minimum wage and employers receive a wage subsidy from the Employment fund.

RMG claimants have to be available for either work in the labour market or for community work, although almost 80 per cent of RMG recipients were exempted from this condition in 1995. Non-exempted recipients agree on an integration contract with the *Service national d'action sociale* within three months of first receiving the benefit. There is little evidence that these pro-

grammes have been successful and chances for RMG receivers to enter the regular labour market seem minimal. Most participants stay for long periods in community service programmes. In some cases, the high level of RMG combined with a rent allowance and free public transport provides little incentive to look for a regular job. Also more stringent work requirements in the regular labour market compared with community service programmes make access for RMG receivers difficult.

Programmes aimed at reducing youth unemployment have been in place since the late 1970s. The main aim is to offer training and work experience to youngsters to improve their chances in the open labour market. The *Division d'auxiliaires temporaires* (DAT) offers job seekers aged less than 30 subsidised employment in the public sector and non-profit organisations. The duration of the programme varies between one and eighteen months and participants can almost immediately end their contract once they have found a regular job. According to the authorities, 70 per cent of participants find regular employment afterwards. Trainee positions are offered to people younger than 30 under the so-called *Contrat de stage-initiation*. The Employment fund will pay the employers' social security contributions plus 25 per cent of the trainee's compensation. The trainee gets priority for recruitment if a suitable position in the enterprise arises. Enterprises can also conclude arrangements with the PES to receive trainees as part of the *Stage de préparation en entreprise*, combining practical training and theoretical study. With a view to improve the transition from school to work, the *Service de formation professionnelle* of the Education Ministry offers a *Cours de formation professionnelle continue* at its training centres.

Faced with low but rising unemployment, in 1993 the Government introduced some special programmes targeted at long-term and older unemployed. With a view to reducing the labour cost of these groups, the Employment fund will pay all the employer's and employee's social security contributions for newly hired people aged 50 or more and for long-term unemployed aged over 30. This subsidy represents some 25 per cent of total labour costs and can – in the case of older workers – continue until the beneficiary reaches the age of 57 and becomes eligible for early retirement.[34] While the budgetary cost of this measure is likely to remain modest given the level of unemployment, a possible adverse impact could be the replacement of younger by older workers.

Increasing the knowledge base, efficiency and innovation capacity

Upgrading skills and competencies

Traditionally language education has an important place in the Luxembourg system and notably in the secondary school system. Hence students can easily continue their study at universities in neighbouring countries since Luxembourg does not have a full university system. However, the predominance of language education makes it more difficult to integrate new subjects such as information technology into the curriculum. Since the beginning of the 1990s, some important reforms have been introduced to improve the performance of the secondary education system. In general education, the reforms have led to a broader curriculum with specialisation only in the last two years of the cycle. Since 1994, changes in examinations have resulted in substantial improvements in exam results. Currently more than 50 per cent of primary school-leavers go on to general secondary education, compared to around 35 per cent in the early 1980s.

The Luxembourg secondary vocational education system has some specific problems. Education is in French and German which can cause problems for both Luxembourg and foreign pupils. The restructuring of the steel industry has led to a substantial decline in employment in the manufacturing sector, and most of the jobs are now in maintenance rather than in production industries. In addition, foreign textbooks have to be adapted for the Luxembourg situation which is a substantial investment, considering the small number of users.

In 1994, the *régime préparatoire* was introduced with the aim of integrating the weakest students into the full-time vocational system. Most of the students in the *régime préparatoire* are foreign and lack sufficient (language) skills to participate in the primary cycle of the vocational system. Most of the classes are French-speaking classes but training prospects for these pupils are still insufficient and not very attractive, since two languages are required for most apprentice schemes. The outcome of the reforms in the primary cycle of the vocational system has not been satisfactory, and further reforms will be implemented once reforms in the intermediate and top levels of the vocational system are completed.

The reform of the secondary vocational system also aimed to harmonise the apprenticeship system. The Luxembourg apprenticeship system is comparable to the German *Dualsystem* and leads to a recognised *Certificat d'aptitude technique et professionnel* (CATP) after three years. For pupils who do not qualify for the

CATP scheme, a more practical three year course is offered leading to a *Certificat de capacité manuelle* (CCM). In 1993, a two-stage apprenticeship course started for pupils who have difficulties assimilating the theoretical and practical aspects at the same speed. After the practical stage, which takes two years, the pupil receives a *Certificat technique et professionnel* (CITP). After the second phase, the student can obtain the CATP. The apprenticeship system is under the control of professional organisations that administer apprentice contracts, training programs and exams at the end of the apprenticeship. The number of apprentices has fallen over the last two decades, reflecting increased participation in general secondary education and a wider choice in the vocational system. The diversification of vocational education has reduced the role of the professional organisations in education.

Product market competition

The Government has pursued an active industrial policy, initially aimed at reducing the economy's dependence on the steel industry, and shifting resources to other sectors, notably to services. Over time, the objectives of the diversification policy have evolved and the aims are now twofold: first to slow down the de-industrialisation process; and second to stop the rise in unemployment. Financial aid and other incentives have been put in place to attract new industries.[35] In addition, the government-owned *Société nationale de crédit et d'investissement* (SNCI) – created in 1977 – is an important instrument in financing investment, mainly in the form of medium and long-term loans. In 1995, the SNCI provided almost LF 6 billion in long-term loans at preferential interest rates to the media sector and the steel industry.

To encourage the reconversion of the steel-producing areas, the Luxembourg government and the French and Belgian regional authorities implemented a cross-border project (*Pôle Européen de Développement,* PED) in 1985. The PED aimed to develop industrial infrastructure with firms starting up in the area enjoying special conditions. Over the last decade, more than 5 000 jobs have been created in the PED, of which 1 243 in Luxembourg. In total, subsidised industries in the manufacturing sector – started up after 1975 – provided work to almost 8 000 people (3½ per cent of total domestic employment) by 1995. As EU regulations restrict this type of industrial policy and general investment grants are no longer permitted, a new industrial diversification law came into force in 1993,

restricting investment grants to regional aid, and introducing three subsidy schemes for small business investment, R&D, energy-saving and environment-protecting investment.

Although the industrial diversification policy has succeeded in creating jobs in the manufacturing sector, it is questionable if this has been done in the most efficient way. The subsidies do not seem to address a clearly identified and serious market failure but are rather directed toward social or political objectives, such as maintaining employment in the steel-producing areas. In the present context of almost full employment, subsidies may be especially harmful because they may have reduced the responsiveness of wages to the deterioration of employment in the steel industry and delayed the adjustment process. In selecting specific investment projects, there is also a risk that government financial assistance substitutes administrative judgement for market judgement as to the most promising industries or growth areas for the economy. If a reduction in subsidies were to result in a scaling back of some sectors, training programmes to promote job mobility would appear to be a more appropriate response, and their cost would probably be lower than providing open-ended subsidies.

Recent policy actions

The authorities feel that there is no need for major new policy initiatives, and find the instruments at their disposal sufficient to reduce unemployment. Hence they have put the emphasis of employment policy on increasing the efficiency of existing active policies, notably programmes targeted at specific groups of unemployed, the reform of the Public Employment Service (PES), and a smoother transition from school to work. Given vigorous employment growth in recent years, the authorities do not feel that measures aimed at stimulating job creation would have a significant impact on unemployment. They are also hesitant to increase labour market flexibility as existing strict social legislation is not seen to have been an impediment to job creation and fear that relaxing job security might lead to the dismantling of the social system.

The social partners and the Government have agreed to strengthen the role of the PES and improve its functioning. The emphasis is on improving the communication between the PES, enterprises and job seekers. The obligation for employers to notify all vacancies to the PES three working days in advance of

any press publication has been more strictly enforced. The PES and the employers' organisations are encouraging enterprises to recruit more people on fixed-term contracts instead of using temporary workers. This may improve recruitment of residents as most temporary workers are cross-border workers.

School leavers, who have not found employment after being registered for three months at the PES are offered a place in one of the youth employment schemes (*Stage-initiation, Stage de préparation, Division d'auxiliaires temporaires*) and benefit payments are made dependent on participation in these schemes. As participation in the schemes does not automatically lead to a regular job, participants continue to receive job offers from the PES, and remain registered as job seekers. The authorities estimate that this requires the creation of 530 additional training places in the public and private sector. To improve the school-to-work transition, school-leavers are encouraged to conclude apprenticeship contracts in sectors with shortages of apprentices. Also apprenticeships have been made more attractive by agreements with employers to guarantee employment after the termination of the apprenticeship. Financial support – similar to that for apprentices – is given to firms that take on trainees from technical secondary education.[36] Furthermore, training courses for low-skilled unemployed have been organised in co-operation with the professional organisation to prepare for CITP and CCM degrees.

The early retirement programme has been made more flexible by the introduction of part-time early retirement (*préretraite progressive*). Under this scheme, workers older than 57 can start working part-time if the employer hires a previously unemployed person for the rest of the time. The advantage of the scheme is that participants can train newly employed persons. Members in the scheme also remain in contact with the labour market so that they can easily re-integrate to full-time status if the labour market improves.

Assessment and scope for further action

Although the overall labour market situation remains rather favourable, following a modest trend increase in registered unemployment the Luxembourg authorities have stepped up corrective initiatives – notably aimed at reducing labour costs for specific groups, increasing training positions, and improving the functioning of the public employment service. These measures do not seem to

have made a significant impact on the number of unemployed. High unemployment in the adjacent regions and increased international mobility of labour limit the potential to reduce unemployment by purely national policy actions. However, the modest scope of the current initiatives – which do not address many of the rigidities in the labour market – also seems to have limited their effectiveness.

The combination of robust employment growth, rising unemployment and falling labour market participation point to structural problems in the labour market: wage flexibility is insufficient, especially at the lower end of the market, and work incentives for certain categories of job seekers are low. To widen the wage structure and bring wages of low-skilled workers more in line with their productivity, wage indexation should be phased out. In addition, minimum wages should be reduced relative to the average wage, possibly by stopping their full indexation and biennial revisions, and the administrative extension of sectoral wage agreements should be relaxed – for instance, through the use of ''opening clauses'' or the creation of wage scales close to the minimum wage – to allow outsiders to price themselves back into the market. If for practical or social reasons wage flexibility cannot be increased sufficiently, more initiatives could be taken to reduce unemployment for low skilled workers, such as reducing non-wage costs through further cuts in the already low taxes and employers' social security contributions at the bottom of the wage scale.

Replacement rates – especially at the minimum wage level – are high and duration periods are in some cases unlimited. This problem is particularly pronounced in Luxembourg, where marginal effective tax rates are high for people in the general assistance scheme (RMG), thus encouraging black-market work and long-term benefit dependency. The Government could improve work incentives by scaling back the generosity of the social security system, thereby freeing up budgetary means to support programmes that promote employment, especially for low-skilled workers, and encouraging persons to shift from a non-active to an active status. To promote labour market participation, the withdrawal rate for additional earnings in the RMG scheme could be reduced from the current 100 per cent, as is the case in some other OECD countries. More importantly, the authorities should review the housing assistance, which is currently conditional on receiving an RMG benefit, so that the loss of such a benefit when income exceeds a certain level, results in a substantial loss in benefit income. If these initiatives were deemed to threaten equity or social goals, these could be best

preserved by the introduction of employment-conditional benefits (or so-called in-work benefits) for low income earners, which could be gradually withdrawn as income rises.

A large number of people have left the labour force and are enrolled in programmes for which no job search is required – like the disability scheme, which has four times as many persons enrolled in it than the unemployment scheme. The authorities should consider tightening access to these schemes by measures concerning disability assessment procedures and encouraging the return of these people into the active labour force. In addition, the disability of those presently in the scheme should be reassessed. Moreover, the working requirement in the RMG scheme should be more strictly applied, notably for older workers for whom there are special programmes to promote re-integration.

The authorities have stepped up active labour market policies aimed at bringing long-term unemployed back into the labour market and improving the quality of the labour force. These programmes have allowed long-term unemployed to gain work experience. In addition, increased flexibility in the labour market and relaxing employment protection legislation may also increase the chances for disadvantaged workers. As a first step, relaxing rules concerning fixed-term contracts may be beneficial to long-term unemployed as experience in other OECD countries shows (notably France and Spain).[37] However, to prevent a wide-spread use of these contracts, it is important to keep protection for regular employees light. The relaxation of Sunday working rules may improve employment prospects for those who are currently outside the active labour force.

Notes

1. Throughout the *Survey*, GDP is based on the 1968 System of National Accounts (SNA).
2. The analysis in this paragraph is based on data from the *Inspection générale de Sécurité sociale*, which may not be fully comparable with data from other sources. The Luxembourg data include employers' social security contributions, but do not include imputed social contributions.
3. Since 1921, the Luxembourg franc has been at par and fully interchangeable with the Belgian franc. Moreover, the Belgian franc is legal tender in Luxembourg (although the Luxembourg franc is not legal tender in Belgium). The National Bank of Belgium (NBB) ensures the liquidity of the Luxembourg banking system and intervenes in the foreign exchange market to keep the Belgian franc – and hence the Luxembourg franc – within the fluctuation margins of the Exchange Rate Mechanism. The foreign reserves of the BLEU are held by the NBB, but the Luxembourg government shares in profit payments of the NBB to the Belgian government.
4. OECD (1995), *OECD Economic Surveys, Belgium/Luxembourg*, Paris.
5. GDP for Luxembourg is underestimated in the 1968 SNA, due to the treatment of imputed banking services as intermediate consumption of resident industries. In the Luxembourg National Accounts version, imputed banking services to non-residents are treated as exports of services. On this basis, total tax revenues were 39.5 per cent of GDP in 1994.
6. The most important special funds are the Employment Fund *(Fonds pour l'emploi*), the Road Fund (*Fonds des routes*) and the Public Investment Funds (*Fonds d'investissements publics*).
7. Until 1996, salaries of employees in nursery and primary education were fully paid by the central government and partly reimbursed by the local authorities. From 1997, only the net contribution of the central government has been included in the budget, implying a reduction in central government spending by about LF 1.2 billion in accounting terms. The medium-term budget norm has not been adjusted downwardly to incorporate these changes.
8. Population, without any further qualification, always refers to the resident population, including foreign nationals.
9. The standardised definition refers to those who actively look for work and are available within two weeks. The national definition of unemployment only refers to those registered at the employment agencies: using this definition, the unemployment rate was 3 per cent in 1995.
10. EUROSTAT (1996), *Labour Force Survey 1995*, Luxembourg. Data from the *Inspection générale de la Sécurité sociale* show a similar trend.

11. EUROSTAT (1996), *op. cit*. According to national data, only 15 per cent of unemployed were registered for more than a year.
12. The official retirement age is 65. However, people can – under certain conditions – take early retirement (*pension de vieillesse anticipée*) at the age of 60. In addition, there are several early retirement programmes (*préretraite ajustement*, for shift workers, *solidarité* and *progressive*) for people older than 57. Miners can take early retirement at the age of 52.
13. Statec (1994), *Recensement de la population au 1er mars 1991*, Luxembourg.
14. Cocher (1993), ''Frontaliers lorrains occupés au G-D de Luxembourg'', *Bulletin de Statec* 5/1993, Luxembourg.
15. All wages and salaries are fully indexed to the consumer price index, with adjustments being made the month after the index has surpassed a certain threshold. Since 1972, wages and salaries rise in steps of 2.5 per cent. The *échelle mobile* can be suspended in case of a deterioration of the economic situation.
16. However, the disposable income at the minimum wage level for a sole family earner with two children is around 80 per cent of the APW's earnings.
17. Apart from Luxembourg the GRT includes the French department of Lorraine, the German state Saar and parts of Rhineland-Palatinate and the Belgian province of Luxembourg. The area has 4.5 million inhabitants and covers 41 000 sq. km.
18. Calculations based on OECD (1995), *The Tax/Benefit Position of Production Workers*, Paris. The tax wedge includes employers' and employees' social security contributions plus income taxes.
19. See Conseil Économique et Social (1995), *Les potentialités et les défis de la grande région transfrontalière*, Luxembourg.
20. See Fehlen, F. and E. Jacquemart (1995), ''La main-d'œuvre frontalière au Luxembourg'', *Cahier Économique* 84, Statec, Luxembourg. They argue that the wage differential may be explained by differences in age, professional status and sector of employment.
21. Cross-border workers are not eligible for a Luxembourg unemployment benefit but receive – in case of unemployment – a benefit in their country of residence.
22. Net replacement rate is defined as net unemployment benefits relative to net earnings.
23. In 1995 only half of registered job seekers received an unemployment benefit while 10 per cent received a social assistance benefit.
24. Legislation concerning RMG is currently being modified. The ten year residency rule will be dropped in the new law.
25. In 1994, 88 per cent of claimants were exempted from being available for work.
26. Additional earnings below 20 per cent of the RMG, are disregarded.
27. Based on calculations by the OECD Directorate for Education, Employment, Labour and Social Affairs.
28. OECD (1994), *The OECD Jobs Study*, Paris.
29. The Employment Minister can accord wage subsidies to firms if working hours have to be reduced due to temporary slack. Wage subsidies can also be granted to firms confronted with

structural problems, provided a restructuring plan exists. Unlike the ordinary unemployment scheme, the part-time unemployment scheme is also open to cross-border workers.

30. See Scarpetta, S. (1996), ''Assessing the role of labour market policies and institutional settings on unemployment: a cross-country study'', *OECD Economic Studies*, No. 26, Paris.

31. Fixed term-contracts are only allowed for carrying out specific tasks of limited duration or as part of active labour market programmes. The maximum duration of a fixed term-contract is two years.

32. See OECD (1993), *Employment Outlook*, Chapter III, Paris.

33. Sunday working is only allowed in bakeries, newsagents and similar establishments. Other shops are permitted to open on Sundays six times a year.

34. For people aged 30 years old and above the maximum duration is two years while for people aged 40 years old and above it is extended to three years.

35. According to the European Commission, state aid to the manufacturing sector – at 4.1 per cent of GDP – was above the European average in the 1990s. European Commission (1995), *Fourth Survey on State Aid in the European Community in the Manufacturing and certain other Sectors*, Brussels.

36. Training to prepare for the CATP, *diplôme de technicien* and *diplôme de fin d'études secondaire technique.*

37. See OECD (1993), Chapter 3, *op. cit.*

Annex

Calendar of main economic events

LUXEMBOURG

1995

May

Following the crossing of the threshold value (*cote d'échéance*) of the *échelle mobile* in April, wages and salaries and all indexed benefits are increased by 2.5 per cent.

The *Comité de coordination tripartite* presents a report on employment policy. The committee concludes that current legislation is sufficient and proposes 35 micro-measures to improve the working of the labour market.

June

A new law on shopping hours comes into force.

The European Investment Bank grants a LF 3 billion loan to the ARBED Group for the modernisation of the steel plant in Belval which should be finished in 1997.

July

Following the advice of the *Comité de coordination tripartite* in May, a law on employment and training is adopted. The main points are the introduction of the *préretraite progressive*; measures concerning vocational training and the functioning of the Public Employment Service; and the installation of a *Comité permanent de l'emploi*.

September

The *Conseil économique et social* publishes a report on the potentials and challenges of the *Grande région transfrontalière* (GRT). A summit meeting between representatives of the GRT is held in Mondorf-les-Bains.

October

The Employment Minister allows Goodyear to introduce Sunday working at its tyre production centre.

The fifth satellite of the ASTRA system is launched. This is the first ASTRA satellite to transmit digital television signals.

The ARBED Group puts an electric furnace in Differdange into use.

1996

January

A wage agreement for the public sector for 1995-99 is approved by Parliament.

The *Comité permanent de l'emploi* is set up to monitor the functioning of the labour market.

April

The sixth satellite of the ASTRA system is launched. It is the second ASTRA satellite to transmit digital television signals.

The *Conseil économique et social* publishes its annual report on economic, financial and social developments.

May

The Prime Minister announces corporate tax reductions – including a steady reduction of the corporate tax rate from 33 per cent in 1996 to 30 per cent in 1999 – during his State of the Nation address.

July

The Government authorises the extension of Luxembourg airport.

September

The Government presents the 1997 Budget to Parliament. Total expenditure is estimated to rise by 4¼ per cent, implying a central government deficit of 1¼ per cent of GDP.

November

The Government presents pension reform plans which will bring public sector pensions in line with those in the private sector.

December

The 1997 Budget is approved by Parliament.

BELGIUM

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Belgium – Selected background statistics**[1]

	Average 1986-95	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Per cent changes from previous year											
Private consumption[2]	2.1	2.3	2.5	2.9	3.9	2.7	2.9	2.3	–0.8	1.3	1.2
Gross fixed capital formation[2]	4.4	4.4	5.6	15.7	11.6	12.0	–5.0	1.8	–5.0	0.3	3.0
Public[2]	–1.4	–4.6	–4.4	6.8	–28.7	–2.9	8.1	7.8	7.8	9.3	–12.9
Residential[2]	7.6	2.2	6.2	22.7	19.2	17.5	–10.4	6.5	3.5	5.5	3.3
Business[2]	4.4	7.2	7.4	14.8	15.3	11.5	–4.1	–0.6	–9.7	–3.2	5.1
GDP[2]	2.2	1.4	2.1	4.9	3.4	3.7	1.6	1.7	–1.4	2.3	1.9
Implicit price deflator											
GDP	3.1	4.7	2.2	1.8	4.6	2.9	3.1	3.6	3.8	2.5	1.4
Private consumption	2.6	2.0	2.4	1.5	3.8	3.3	3.1	2.1	3.0	3.2	1.6
Exports of goods and services	–0.6	–8.4	–3.5	2.9	6.9	–1.5	–0.5	–1.0	–1.2	0.3	–0.1
Imports of goods and services	–1.3	–12.2	–4.3	2.6	6.3	–1.1	–0.6	–2.7	–2.6	0.9	0.5
Industrial production	1.5	1.0	2.2	5.5	3.8	3.7	–2.0	–0.3	–4.9	1.6	4.3
Employment	0.4	0.6	0.4	1.5	1.6	1.4	0.1	–0.4	–1.1	–0.7	0.4
Compensation of employees	4.6	4.1	3.1	3.2	6.5	7.6	7.9	5.3	2.5	3.0	2.8
Productivity (GDP[2]/employment)	1.8	0.7	1.7	3.4	1.8	2.2	1.4	2.1	–0.3	3.1	1.5
Unit labour costs (comp. of employees/GDP[2])	2.4	2.7	1.0	–1.7	2.9	3.8	6.2	3.5	3.9	0.7	0.9
B. Percentage ratios											
Gross fixed capital formation as % of GDP[2]	18.1	15.4	15.9	17.5	18.9	20.5	19.1	19.1	18.4	18.1	18.2
Stockbuilding as % of GDP[2]	0.2	–0.4	0.3	0.5	0.4	0.0	0.1	0.2	0.1	0.3	0.4
Foreign balance as % of GDP[2]	3.3	4.4	3.1	3.5	2.7	2.7	3.0	2.7	2.8	3.7	4.0
Compensation of employees as % of GDP	53.2	55.0	54.3	52.5	51.6	52.1	53.7	53.6	53.6	52.7	52.4
Direct taxes as % of household income	13.1	14.3	14.2	13.6	12.6	13.0	12.3	12.4	12.1	13.0	13.3
Household saving as % of disposable income	17.5	15.9	14.4	15.6	16.5	17.0	18.9	19.9	20.9	18.5	17.8
Unemployment as % of civilian labour force	11.0	11.8	11.5	10.4	9.4	8.8	9.4	10.4	12.1	13.1	13.0
C. Other indicator											
Current balance (BLEU) (billion US$)	7.6	–2.0	1.7	2.2	2.1	7.0	9.9	11.9	15.9	12.5	15.1

1. All variables include statistical discrepancies.
2. At constant 1990 prices.

Source: National Accounts Institute and OECD Secretariat.

Table B. **Belgium – Gross domestic product**[1]

Billion francs

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Current prices									
Private consumption	3 241.1	3 402.8	3 553.8	3 831.9	4 065.2	4 313.0	4 507.9	4 602.1	4 809.7	4 946.8
Public consumption	846.7	851.8	852.9	878.8	912.9	982.1	1 019.3	1 081.1	1 129.1	1 172.8
Gross fixed capital formation	783.6	836.8	987.7	1 150.0	1 324.2	1 284.3	1 345.2	1 299.3	1 332.2	1 395.7
Change in stocks	–29.6	11.6	19.8	23.0	–2.0	4.6	3.3	–3.4	24.1	33.8
Total domestic demand	4 841.8	5 102.9	5 414.3	5 883.6	6 300.2	6 583.9	6 875.7	6 979.1	7 295.1	7 549.1
Exports of goods and services	3 522.9	3 608.8	4 028.8	4 629.8	4 733.2	4 854.5	4 980.3	5 011.7	5 490.6	5 764.0
less: Imports of goods and services	3 329.4	3 455.6	3 830.4	4 441.8	4 558.6	4 658.3	4 713.2	4 674.2	5 107.6	5 377.1
Gross domestic product at market prices	5 035.3	5 256.1	5 612.7	6 071.6	6 474.8	6 780.1	7 142.8	7 316.6	7 678.1	7 936.0
	1990 prices									
Private consumption	3 613.7	3 704.2	3 810.1	3 958.7	4 065.2	4 183.6	4 280.9	4 244.7	4 300.7	4 352.4
Public consumption	932.2	934.7	926.6	916.9	912.9	932.9	933.9	947.1	960.5	969.6
Gross fixed capital formation	866.8	915.7	1 059.2	1 181.9	1 324.2	1 257.6	1 279.7	1 215.4	1 219.0	1 255.1
Change in stocks	–24.1	19.2	29.2	21.9	–2.0	6.9	11.5	3.7	23.0	29.1
Total domestic demand	5 388.6	5 573.8	5 825.1	6 079.4	6 300.2	6 380.9	6 506.0	6 410.8	6 503.2	6 606.3
Exports of goods and services	3 680.1	3 905.8	4 239.0	4 558.4	4 733.2	4 881.2	5 056.3	5 149.5	5 621.9	5 905.8
less: Imports of goods and services	3 434.9	3 726.1	4 027.3	4 392.9	4 558.6	4 685.5	4 874.6	4 963.0	5 373.3	5 629.9
Gross domestic product at market prices	5 633.8	5 753.5	6 036.8	6 244.9	6 474.8	6 576.6	6 687.7	6 597.3	6 751.8	6 882.2

1. Including statistical discrepancies.

Source: National Accounts Institute, *Comptes nationaux 1995* and OECD, *National Accounts*.

Table C. **Belgium – Income and expenditure of households and private non-profit institutions**

Billion francs

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Compensation of employees	2 767.7	2 854.6	2 945.7	3 135.8	3 373.4	3 639.5	3 830.7	3 925.0	4 044.2	4 157.6
Income from firms received by individuals	913.9	942.5	1 038.9	1 074.6	1 162.1	1 197.1	1 254.5	1 288.0	1 369.1	1 412.2
Household property income [1]	663.6	679.1	734.9	844.2	941.8	1 035.6	1 134.6	1 196.5	1 183.9	1 165.1
Current transfers from government	1 372.2	1 439.6	1 492.7	1 581.1	1 676.1	1 818.9	1 944.7	2 024.4	2 086.3	2 164.4
Current transfers from the rest of the world	33.2	39.4	50.7	59.0	66.7	69.9	71.1	77.8	76.9	75.6
Household income	5 750.6	5 955.2	6 262.9	6 694.7	7 220.1	7 761.0	8 235.6	8 511.7	8 760.4	8 974.9
less:										
Direct taxes	822.1	844.4	853.3	846.2	938.0	957.7	1 020.2	1 029.1	1 143.1	1 196.7
Social security contributions by wage-earners and self-employed	789.0	836.9	870.9	920.2	987.6	1 068.8	1 141.5	1 184.9	1 211.9	1 244.4
Current transfers to the rest of the world	42.8	43.7	49.2	57.0	89.0	84.9	93.8	104.7	110.4	105.9
Other current transfers	240.8	255.5	280.7	284.1	306.5	329.4	351.2	378.2	392.2	413.5
Disposable income	3 855.9	3 974.7	4 208.8	4 587.3	4 899.1	5 320.1	5 628.9	5 814.8	5 902.8	6 014.4
Household savings	614.8	571.9	655.0	755.4	833.8	1 007.1	1 121.0	1 212.7	1 093.1	1 067.6

1. Including statistical discrepancy.

Source: National Accounts Institute, *Comptes nationaux 1995* and OECD, *National Accounts*.

Table D. **Belgium – Income and expenditure of enterprises**

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
	Billion francs									
1. Gross operating surplus excluding subsidies	657	709	841	984	990	942	1 000	1 004	1 099	1 207
2. Subsidies	184	173	177	162	192	219	211	202	206	226
3. **Gross operating surplus** (1 + 2)	841	882	1 018	1 146	1 183	1 162	1 211	1 205	1 306	1 433
4. Net property income payable	141	138	179	196	263	308	329	294	287	307
5. **Gross primary income** (3 – 4)	700	744	839	950	919	853	882	912	1 019	1 126
6. Current transfers paid to other sectors	126	127	139	147	126	133	136	157	178	208
7. **Disposable income** (5 – 6)	574	616	700	803	794	720	745	754	840	918
8. Capital transfers (net)	39	29	35	30	36	43	46	57	35	63
9. **Capital resources** (7 + 8)	613	645	735	833	829	762	791	812	876	982
10. Gross capital formation	420	497	591	721	805	805	813	732	744	808
11. Other[1]	11	12	17	20	18	20	16	22	30	39
12. **Net lending** (9 – 10 – 11)	182	136	127	92	7	–62	–37	58	102	134
	Per cent of GDP									
Gross operating surplus (including subsidies)	16.7	16.8	18.1	18.9	18.3	17.1	17.0	16.5	17.0	18.1
Disposable income	11.4	11.7	12.5	13.2	12.3	10.6	10.4	10.3	10.9	11.6
Capital resources	12.2	12.3	13.1	13.7	12.8	11.2	11.1	11.1	11.4	12.4
Gross capital formation	8.3	9.5	10.5	11.9	12.4	11.9	11.4	10.0	9.7	10.2
Net lending	3.6	2.6	2.3	1.5	0.1	–0.9	–0.5	0.8	1.3	1.7

1. Change in mathematical retirement pension reserves and net purchases of land and intangible assets.
Source: National Accounts Institute, *Comptes nationaux 1995*.

Table E. **Belgium – Government revenue and expenditure**

Billion francs

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Current revenue	2 028.9	2 150.0	2 224.7	2 288.4	2 464.9	2 642.7	2 728.1	2 843.2	3 111.6	3 310.2
Income from property and firms	72.1	62.4	64.8	75.7	85.1	93.6	88.8	88.8	76.7	79.1
less: Interest on public debt	–563.0	–557.9	–568.5	–629.4	–686.1	–692.0	–772.0	–790.3	–773.6	–722.9
Indirect taxes	594.7	645.1	675.5	737.5	786.1	823.1	865.6	908.0	980.3	983.2
Direct taxes on households	822.1	844.4	853.3	846.2	938.0	957.7	1 020.2	1 029.1	1 143.1	1 196.7
Social security contributions by wage-earners and self-employed	789.0	836.9	870.9	920.2	987.6	1 068.8	1 141.5	1 184.9	1 211.9	1 244.4
Direct taxes on companies	140.1	143.7	157.1	173.9	159.3	164.0	155.1	180.1	223.2	255.1
Other current transfers	173.8	175.2	171.7	164.4	195.0	227.5	228.9	242.6	249.9	274.6
Current expenditure	2 377.6	2 440.8	2 490.9	2 587.0	2 734.4	2 979.7	3 118.0	3 249.8	3 379.2	3 506.6
Public consumption	846.7	851.8	852.9	878.8	912.9	982.1	1 019.3	1 081.1	1 129.1	1 172.8
Subsidies	183.7	173.5	177.2	161.9	192.3	219.4	211.0	201.7	206.5	226.3
Social security transfers to wage-earners and self-employed	1 024.9	1 067.2	1 087.4	1 140.1	1 206.7	1 312.9	1 402.4	1 453.5	1 493.5	1 544.6
Other current transfers (net) to households	118.6	124.7	134.0	152.8	158.6	169.4	178.1	184.9	202.8	206.2
Other current transfers	203.7	223.7	239.4	253.4	263.8	296.0	307.1	328.6	347.3	356.7
Savings of general government	–348.7	–290.9	–266.2	–298.6	–269.4	–337.0	–390.0	–406.5	–267.6	–196.4

Source: National Accounts Institute, *Comptes nationaux 1995* and OECD, *National Accounts*.

Table F. **Belgium – Area breakdown of foreign trade for the BLEU**

Million US$

	1990	1991	1992	1993 [1]	1994	1995
Exports, fob	118 034	118 317	122 987	117 687	134 301	163 047
OECD	104 899	104 305	108 106	101 875	116 575	140 126
EU	92 737	92 718	96 059	88 434	100 374	121 707
of which:						
Germany	25 164	28 114	28 150	24 650	27 848	34 917
France	23 877	22 561	23 763	22 172	25 288	29 400
Netherlands	16 088	16 202	16 605	15 339	17 563	21 438
United Kingdom	10 239	9 161	9 621	9 850	11 277	13 065
Italy	7 729	7 183	7 250	6 444	6 876	8 934
USA	5 096	4 431	4 773	5 655	6 855	6 652
Other OECD	7 066	7 155	7 275	7 787	9 345	11 768
Non-OECD	12 221	13 031	14 045	15 233	17 082	22 141
Ex-COMECON	1 080	1 514	1 780	1 272	1 562	1 860
OPEC	2 124	2 504	2 624	2 510	2 146	2 729
Other	9 017	9 012	9 641	11 451	13 374	17 552
Unspecified	914	981	836	579	645	779
Imports, cif	120 043	121 036	124 967	105 427	122 007	148 658
OECD	105 608	106 739	110 354	93 145	106 508	131 219
EU	92 058	92 570	96 045	80 129	91 110	111 233
of which:						
Germany	28 803	28 458	29 872	22 720	24 089	30 356
France	19 021	19 090	20 602	17 090	19 410	22 729
Netherlands	21 008	20 868	21 884	18 430	21 614	26 415
United Kingdom	9 932	10 127	9 624	9 941	11 551	13 263
Italy	5 413	5 485	5 665	4 680	5 184	6 290
USA	5 452	5 807	5 402	5 601	6 468	8 748
Other OECD	8 098	8 361	8 907	7 415	8 930	11 239
Non-OECD	14 373	14 220	14 548	12 161	14 457	17 173
Ex-COMECON	2 046	2 013	2 087	742	1 206	1 257
OPEC	3 292	3 616	3 309	1 212	1 582	1 462
Other	9 035	8 591	9 153	10 207	11 669	14 454
Unspecified	62	77	65	121	1 042	265

1. As a result of the abolition of customs frontiers within the EU on 1st January 1993, data on intra-EU trade are no longer derived from customs declarations but based on INTRASTAT.

Source: OECD, *Foreign Trade Statistics*, Series A.

Table G. **Belgium – Commodity breakdown of foreign trade for the BLEU**

Million US$

	1989	1990	1991	1992	1993 [1]	1994
Exports, fob	100 741	118 002	118 222	122 411	120 331	134 053
SITC sections						
0. Food and live animals	8 581	9 848	10 862	11 892	11 381	12 282
1. Beverages and tobacco	691	786	813	870	1 069	1 205
2. Crude material, inedible, except fuels	2 828	2 936	2 667	2 777	2 646	3 116
3. Mineral fuels, lubricants and related materials	3 486	4 125	4 624	4 243	4 095	4 073
4. Animal and vegetable oils and fats	477	487	492	483	403	531
5. Chemicals	13 889	16 477	16 707	18 012	18 774	22 838
6. Manufactured goods classified chiefly by material	33 308	36 732	35 021	34 711	33 043	36 579
7. Machinery and transport equipment	25 663	32 268	32 017	33 174	33 012	37 788
8. Miscellaneous manufactured articles	7 565	9 701	10 095	11 131	10 936	11 273
9. Other	4 253	4 641	4 923	5 117	4 971	4 368
Imports, cif	99 340	119 725	121 038	124 656	110 815	122 917
SITC sections						
0. Food and live animals	8 027	9 425	10 193	10 867	9 808	10 884
1. Beverages and tobacco	1 032	1 304	1 342	1 545	1 581	1 597
2. Crude material, inedible, except fuels	6 738	6 856	6 278	6 260	5 035	6 325
3. Mineral fuels, lubricants and related materials	7 637	9 623	10 171	9 489	8 309	8 414
4. Animal and vegetable oils and fats	383	406	446	457	438	602
5. Chemicals	11 514	13 701	13 980	14 892	14 484	16 268
6. Manufactured goods classified chiefly by material	24 497	28 093	26 960	27 244	24 701	27 789
7. Machinery and transport equipment	24 183	30 610	31 576	32 017	27 953	31 444
8. Miscellaneous manufactured articles	9 751	12 635	13 225	14 444	12 665	13 307
9. Other	5 578	7 071	6 866	7 441	5 843	6 288

1. As a result of the abolition of customs frontiers within the EU on 1st January 1993, data on intra-EU trade are no longer derived from customs declarations but based on INTRASTAT.

Source: OECD, *Foreign Trade Statistics*, Series C.

Table H. **Belgium – BLEU balance of payments**

Million US$

	1986	1987	1988	1989	1990 [1]	1991	1992	1993	1994	1995 [1]
Exports, fob [2]	64 531	80 112	89 887	94 290	112 692	110 218	119 821	109 045	122 366	157 793
Imports, fob [2]	68 233	80 744	89 370	94 349	113 862	110 144	118 590	105 833	115 564	148 978
Trade balance	–3 702	–632	517	–59	–1 170	74	1 231	3 212	6 802	8 815
Services, net	2 932	3 478	3 461	4 346	5 109	6 643	6 602	9 642	10 873	..
Balance on goods and services	–1 045	3 232	4 008	4 026	9 217	12 082	14 596	18 725	15 975	18 332
Private transfers, net	–215	–115	46	51	–670	–439	–706	–773	–1 005	–532
Official transfers, net	–759	–1 384	–1 857	–1 995	–1 562	–1 774	–2 013	–2 034	–2 504	–2 709
Current balance	–2 019	1 733	2 197	2 082	6 985	9 869	11 877	15 918	12 466	15 091
Long-term capital (excluding special transactions)	–3 762	–1 344	–2 817	–2 216	8 028	10 863	9 192	20 807	–11 738	..
Private	–3 480	–597	–2 537	–1 952	7 423	8 995	12 692	10 348	–6 535	..
Official	–282	–747	–280	–264	604	1 868	–3 500	10 458	–5 202	..
Basic balance	–658	1 425	710	1 335	11 668	15 765	15 778	32 148	509	..
Non-monetary short-term capital										
Private	–770	246	1 221	–673	2 196	3 284	–121	–5 588	–1 189	..
Official	1 222	977	2 657	2 135	..	..	..	..	..	..
Errors and omissions	206	56	–139	–614	–1 574	–717	1 835	–634	–2 277	..
Balance on non-monetary transactions	0	2 704	4 449	2 183	12 290	18 332	17 492	25 926	–2 957	..
Memorandum item:										
Exchange rate (francs per US$)	44.69	37.34	36.77	39.40	33.42	34.16	32.15	34.55	33.46	29.50

1. Break in series.
2. Including commission processing and non-monetary gold.

Source: OECD Secretariat.

Table I. **Belgium – Structure of output and performance indicators**

1. Structure of output (constant prices)

	1980	1985	1990	1992	1993	1994	1995	1980	1985	1990	1992	1993	1994	1995
	Share of GDP							Share of total employment						
Agriculture, hunting, forestry and fishing	2.0	2.2	1.9	2.2	2.3	2.1	2.1	3.1	3.1	2.7	2.5	2.5	..	..
Mining and quarrying	0.3	0.2	0.2	0.3	0.2	0.2	0.3	..	..	..	..	..	..	..
Manufacturing	20.7	22.8	22.2	21.3	20.9	21.4	21.6	24.9	22.5	20.8	20.0	19.3	..	..
of which:														
Food products	4.4	4.5	4.0	4.0	4.0	4.0	4.1	3.1	3.0	2.8	2.7	2.7	..	..
Textiles	1.6	1.4	1.6	1.5	1.5	1.5	1.4	1.9	1.8	1.5	1.3	1.4	..	..
Paper	1.1	1.2	1.4	1.3	1.4	1.4	1.4	1.6	1.6	1.6	1.5	1.5	..	..
Chemicals	2.0	3.1	3.3	3.5	3.4	3.6	3.8	2.0	2.0	2.0	2.0	1.9	..	..
Metals	1.5	1.7	1.7	1.5	1.5	1.5	1.6	2.2	1.8	1.4	1.3	1.2	..	..
Metal products, machinery and equipment	7.0	8.1	7.0	6.2	6.0	6.2	6.1	8.5	7.6	7.2	6.9	6.6	..	..
Electricity, gas and water	2.5	2.6	2.5	2.6	2.6	2.6	2.7	1.7	1.6	1.0	1.0	1.0	..	..
Construction	6.7	4.6	5.3	5.4	5.3	5.3	5.2	7.8	5.7	6.3	6.5	6.8	..	..
Services	48.9	50.1	50.6	51.7	52.6	52.3	52.4	40.2	43.3	46.4	47.5	48.3	..	..
Total market sector[1]	81.1	82.5	82.8	83.4	83.9	83.9	84.2	77.6	76.2	77.1	77.5	77.9	..	..
Non-market sector[1]	13.2	13.2	11.6	11.7	12.0	12.0	11.9	17.3	18.3	17.5	17.2	17.3	..	..

1. The total market sector and non-market sector percentages do not add up to 100 due to an ''undefined'' category.

Source: National Accounts Institute, National Bank of Belgium and OECD, *National Accounts.*

Table I. **Belgium – Structure of output and performance indicators** *(cont.)*

2. Economic performance (constant prices)

	1980	1985	1990	1992	1993	1994	1995	Average 1980-85	1990	1992	1993	1994	1995
	Share of total investment							Productivity growth					
Agriculture, hunting, forestry and fishing	2.1	2.5	1.9	1.8	1.6	1.7	1.6	4.2	–1.9	14.1	6.6	. .	. .
Mining and quarrying	0.4	0.5	0.7	0.3	0.3	0.3	0.3	. .	. .	. .	. .	. .	. .
Manufacturing	13.9	20.6	27.7	25.6	19.6	19.0	19.9	6.1	1.7	1.9	1.3	. .	. .
of which:													
Food products	2.2	2.6	3.4	3.9	3.0	2.7	2.5	2.2	–0.8	0.7	2.5	. .	. .
Textiles	0.8	1.8	1.7	1.4	1.2	1.4	1.2	0.4	8.7	8.7	–3.5	. .	. .
Paper	1.0	1.5	2.1	2.6	1.8	2.6	2.2	2.2	1.0	3.3	8.1	. .	. .
Chemicals	2.9	4.2	10.2	7.7	6.6	5.9	6.8	11.3	–0.9	6.0	–0.2	. .	. .
Metals	1.7	2.2	1.7	2.0	1.5	1.5	1.5	8.0	3.6	–0.1	5.1	. .	. .
Metal products, machinery and equipment	3.8	6.4	6.2	6.1	3.8	3.2	4.0	6.8	1.8	–1.5	–0.3	. .	. .
Electricity, gas and water	5.3	6.3	3.2	5.3	5.6	5.1	5.2	2.8	13.4	3.5	1.3	. .	. .
Construction	1.6	1.7	2.2	1.9	1.6	1.7	1.8	0.0	4.1	3.1	–5.4	. .	. .
Services	59.6	55.9	58.7	58.2	63.5	64.0	64.2	0.3	0.1	1.6	–0.3	. .	. .
Total market sector	82.9	87.5	94.4	93.1	92.2	91.8	93.0	2.1	0.9	2.1	–0.1	. .	. .
Non-market sector	17.1	12.5	5.6	6.9	7.8	8.2	7.0	0.3	–0.4	3.1	1.7	. .	. .

Source: National Accounts Institute, National Bank of Belgium and OECD, *National Accounts.*

Table I. **Belgium – Structure of output and performance indicators** *(cont.)*

3. Other indicators (current prices)

	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
R&D as a per cent of GDP in manufacturing sector	4.5	. .	4.9	5.0	5.3	5.1	4.8	. .	5.1	. .	. .	. .	. .
Total R&D expenditure as a per cent of total GDP	1.6	1.6	1.7	1.7	1.7	1.6	1.7	. .	1.7	. .	1.6	. .	. .
Government funded R&D as a per cent of total	33.4	31.8	31.6	28.7	27.6	26.7	32.0	. .	31.3	. .	32.5	. .	. .
Breakdown of employed workforce by size of establishment:													
1 to 9 employees	14.6	14.6	14.7	14.9	15.1	16.0	16.1	16.0	16.1	16.3	16.4	16.7	16.7
10 to 49 employees	20.2	19.8	20.0	20.4	20.8	21.5	21.8	22.1	22.4	22.5	23.3	23.3	23.3
50 to 199 employees	21.1	20.9	20.9	21.0	21.6	21.6	21.4	21.3	21.7	21.8	22.4	22.5	22.7
200 to 499 employees	15.0	14.9	14.8	14.4	14.7	14.6	14.9	15.0	14.5	14.8	14.6	14.3	14.2
500 to 999 employees	9.8	10.1	9.5	9.6	9.6	9.4	9.4	9.2	9.3	9.3	9.3	9.5	9.3
1 000 employees or more	19.4	19.8	20.2	19.8	18.2	17.0	16.5	16.4	15.9	15.4	14.0	13.7	13.8
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Workforce (thousands)	2 818	2 824	2 845	2 863	2 889	3 003	3 082	3 145	3 154	3 159	3 156	3 142	3 188

Source: National Social Security Office and OECD, *Main Science and Technology Indicators*.

Table J. **Belgium – Labour market indicators**

	1980	1985	1990	1992	1993	1994	1995
A. Trend							
Standardised unemployment rate	8.8	11.4	7.2	7.7	8.6	9.6	9.5
Unemployment rate	7.9	12.3	8.7	10.3	12.0	12.9	12.9
Male	4.6	8.8	5.9	7.4	8.9	9.8	9.7
Female	13.5	17.7	12.8	14.2	16.1	17.1	17.1
Youth (15-24 years old)	. .	23.6	14.5	13.2	18.4	21.8	21.5
Share of long-term unemployment [1]	. .	68.4	67.3	59.0	53.0	58.3	62.4
Unfilled vacancies (thousands)	5.9	18.3	18.8	16.8	14.6	19.0	19.7
B. Structural and institutional features							
Labour force (per cent change)	0.3	–0.5	0.8	0.7	0.9	0.4	0.4
Participation rate [2]	62.3	61.3	61.9	62.7	63.1	63.3	63.4
Male	78.9	74.5	72.7	72.6	72.4	72.4	72.5
Female	47.0	49.3	52.4	54.1	55.1	55.4	55.8
Employment as a per cent of population aged 15 to 64	57.3	53.7	56.4	56.2	55.5	55.0	55.1
Employers, self-employed and family workers (as a per cent of total employment)	16.4	17.8	17.9	18.3	18.8	19.1	19.2
Employees (as a per cent of total employment)	83.6	82.2	82.1	81.7	81.2	80.9	80.8
Civilian employment by sector (as a per cent of total)							
Agriculture	3.2	3.1	2.7	2.6	2.5	2.5	. .
Industry	34.7	30.2	28.3	27.7	27.2	27.0	. .
Services	62.1	66.7	68.9	69.7	70.3	70.6	. .
of which: General government	19.2	20.7	20.0	19.5	19.4	19.1	. .
Total	100.0	100.0	100.0	100.0	100.0	100.0	. .
Non-wage labour costs [3]	20.3	23.4	26.4	26.6	26.9	26.6	26.7
Unemployment insurance benefits [4]	33.5	28.5	32.6	30.4	28.2	. .	. .

1. People who have been looking for a job for one year or more as a percentage of total unemployment.
2. Labour force as a percentage of the corresponding population aged between 15 and 64 years.
3. Employers' social security contributions as a percentage of total wages.
4. Average unemployment benefit as a percentage of compensation per employee.

Source: National Institute of Statistics, *Annuaire statistique de la Belgique 1994*; Ministry of Finance, *Note de conjoncture* 1996/6/131; OECD, *Labour Force Statistics* (Parts II and III) and *Main Economic Indicators.*

Table K. **Belgium – Public sector**

	1975	1980	1985	1990	1992	1993	1994	1995
Budgetary indicators: general government accounts (as a per cent of GDP)								
Primary receipts [1]	45.4	48.7	52.1	47.7	48.2	48.9	50.0	50.4
Primary total expenditure [1]	47.5	52.2	51.2	43.7	45.4	46.3	45.7	45.9
Primary balance [1]	–2.2	–3.5	0.8	4.0	2.8	2.6	4.3	4.5
Net interest payments	2.9	5.4	9.8	9.7	10.0	10.0	9.4	8.7
Net lending	–5.0	–8.9	–9.0	–5.6	–7.2	–7.5	–5.1	–4.1
Structure of expenditure and taxes (as a per cent of GDP)								
General-government current expenditure	46.3	53.7	59.0	52.8	54.5	55.2	54.1	53.3
of which:								
Transfers	23.0	25.8	27.1	25.2	26.4	26.9	26.6	26.6
Subsidies	3.4	4.0	4.0	3.0	3.0	2.8	2.7	2.9
Tax receipts	41.8	44.4	47.7	44.4	44.6	45.3	46.3	. .
Personal income tax	13.3	15.6	16.7	14.1	13.9	13.6	14.4	. .
Corporate taxes	3.0	2.5	2.6	2.4	2.1	2.4	2.8	. .
Social security contributions	13.3	13.5	15.4	15.0	15.7	16.0	15.4	. .
Consumption taxes	11.0	11.6	12.0	11.6	11.7	12.0	12.4	. .
of which: Value added tax	6.6	7.5	7.4	7.2	7.0	7.0	7.2	. .
Other indicators								
Income tax elasticity	1.8	0.5	1.5	1.4	1.0	0.0	3.8	. .
Income tax as per cent of total tax	39.3	41.0	40.7	37.4	36.2	35.7	37.4	. .
Gross general government debt (as a per cent of GDP)	58.7	78.7	123.1	129.7	130.7	137.0	135.0	133.7
Net general government debt (as a per cent of GDP)	48.9	69.6	112.6	118.2	121.3	127.4	126.7	127.7

	Prior to		After
Tax rates (per cent)			
Personal income tax rates			
Top rate	72	1st January 1989	55 [2]
Lower rate	17	1st January 1989	25 [2]
Number of brackets	13	1st January 1989	7
Corporate tax rates	41	1st January 1991	39 [2]
Standard VAT rate	20.5	1st January 1996	21

1. Excluding interest charges.
2. Excluding the solidarity tax surcharge (*contribution complémentaire de crise*) which has increased the tax rates by 3 per cent since 1st August 1993 (to 56.65, 35.75 and 40.17 per cent respectively).

Source: National Bank of Belgium; OECD, *National Accounts, Revenue Statistics 1965-1995* and OECD Secretariat.

Table L. **Belgium – Financial markets**

1. Sector size and structure of financial flows

	1975	1980	1990	1991	1992	1993	1994	1995
Sector size								
Sectoral employment[1]/total employment	. .	1.8	2.1	2.0	2.0	2.0	2.0	2.0
Non-financial agents' accumulation of financial liabilities/GDP	16.7	21.2	18.7	21.4	17.8	16.0	10.7	10.1
Non-financial agents' accumulation of financial assets/GDP	17.5	16.8	18.6	22.6	19.8	21.7	14.7	14.9
Stock-market capitalisation/GDP	16.0	9.2	31.6	33.2	30.2	38.5	34.9	37.6
Density of banking network[2]	35.9	39.0	36.3	35.5	34.9	34.8	. .	. .
Density of banking network: credit institutions[3]	. .	. .	90.2	87.4	80.0	78.1	76.9	75.6
Structure of financial flows								
Share of intermediated financing in total financing[4]	. .	. .	83.2	60.7	64.6	22.8	98.5	33.8
Financial institutions' share of financial assets	. .	. .	60.7	61.5	36.8	51.6	46.0	13.6
Share of securities issues in financial flows								
of non-financial agents	37.1	35.2	64.3	61.4	74.2	31.7	76.7	112.4
Structure of private non-financial sector's portfolio:[5]								
Deposits	63.9	36.6	33.1	17.7	40.1	48.6	34.5	27.2
Bonds and bills	35.4	56.4	65.3	62.0	45.0	42.2	23.7	39.6
Equities	4.6	0.3	3.8	5.7	0.4	4.5	16.7	13.8
Non-financial corporate financial structure:								
Equity	. .	. .	34.7	35.5	35.9	36.5	36.5	37.4
Short-term debt	. .	. .	40.3	40.1	40.1	38.9	58.3	38.2
Long-term debt	. .	. .	21.1	20.5	20.1	20.3	20.9	20.2

1. Only financial institutions.
2. Number of deposit banks' branches and head offices per 100 000 population.
3. Number of bank branches and head offices per 100 000 population.
4. Share of financial institutions in total external financing of non-financial agents. The 1993 figure was affected by a sharp increase in the issue of Belgian investment funds (CIUs).
5. The private non-financial sector includes corporations as well as households and non-profit institutions. The total differs from 100 because certain items, such as loans and trade credit, are excluded.

Source: National Bank of Belgium.

Table L. **Belgium – Financial markets** *(cont.)*

2. Internationalisation and efficiency of markets

	1975	1980	1990	1991	1992	1993	1994	1995
Internationalisation of markets								
Foreign business of the banking sector:[1]								
Assets	38.4	46.8	48.6	49.4	48.8	35.2	35.2	35.2
Liabilities	43.6	56.1	59.6	57.9	57.3	40.8	40.2	40.2
International banking network:								
Foreign banks in Belgium[2]	40	51	63	67	73	80	78	78
Belgian bank branches abroad	. .	. .	52[3]	58[3]	64	69	. .	. .
Share of cross-border transactions:								
Net purchases of foreign securities by residents[4]	25.6	11.1	16.7	27.6	27.6	76.3	45.3	60.4
Net purchases of domestic securities by non-residents[5]	9.6	19.4	18.9	39.1	25.3	49.5	42.5	33.9
Efficiency of markets								
Cost of bank intermediation[6]	3.1	2.5	1.7	1.8	1.9	1.9	1.7	1.8
Bank productivity[7]	73.6	74.6	69.3	65.7	64.5	67.6	71.6	66.9
Interest margins[8]	2.5	2.0	1.3	1.3	1.4	1.2	1.3	1.2

1. As a percentage of deposit banks' balance sheets.
2. Number of branches and subsidiaries.
3. Number of deposit banks' branches and head offices.
4. Purchases of foreign shares and other securities, equity participations in foreign entreprises as a percentage of total purchases of domestic and foreign securities by the private non-financial sector.
5. Purchases of Belgian shares, foreign equity participations in Belgian entreprises and other purchases of securities issued by residents as a percentage of domestic securities issues.
6. Gross benefit margins as a percentage of the annual average balance sheet of deposit banks.
7. Operating costs as a percentage of the gross benefit margins of deposit banks.
8. Difference between interest receipts and interest payments divided by the annual average balance sheet of deposit banks.

Source: National Bank of Belgium.

LUXEMBOURG

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Luxembourg – Selected background statistics**[1]

	Average 1986-95	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
A. Annual percentage change											
Private consumption[2]	3.8	5.7	4.6	4.6	5.1	5.7	6.4	–0.9	1.7	2.4	2.4
Gross fixed capital formation[2]	11.3	31.1	17.9	15.2	7.0	2.7	31.5	–9.0	28.4	–14.8	3.5
GDP[2]	6.0	7.7	2.3	10.4	9.8	2.1	6.2	4.4	8.7	4.2	3.7
GDP price deflator	2.6	2.7	1.0	0.7	3.5	3.4	1.5	4.3	0.7	5.4	3.1
Industrial production	2.1	1.8	–0.7	8.9	7.9	–0.4	–0.1	–1.1	–2.2	6.1	0.7
Employment	2.9	2.6	2.6	3.0	3.5	4.1	4.1	2.5	1.8	2.3	2.5
Compensation of employees	9.1	9.0	7.3	7.1	12.1	10.5	11.4	8.2	7.1	. .	. .
Productivity (GDP[2]/employment)	3.0	5.0	–0.3	7.2	6.1	–1.9	2.0	1.9	6.8	1.9	1.3
Unit labour costs (compensation/GDP[2])	2.5	1.1	4.9	–3.0	2.1	8.2	4.9	3.6	–1.5	. .	. .
B. Percentage ratios											
Gross fixed capital formation as % of GDP[2]	24.1	19.5	22.5	23.4	22.8	23.0	28.4	24.8	29.3	23.9	23.9
Stockbuilding as % of GDP[2]	–2.6	–1.6	–3.5	–3.7	–0.9	–2.2	–5.4	–3.6	–3.3	–0.8	–0.8
Foreign balance as % of GDP[2]	7.3	5.2	2.4	5.7	6.9	5.9	3.7	9.2	8.6	12.6	13.2
Compensation of employees as % of GDP	56.2	54.8	57.0	54.9	54.2	56.7	58.6	58.2	56.9	55.5	55.6
Number of unemployed	3 010	2 300	2 700	2 500	2 300	2 100	2 300	2 700	3 500	4 600	5 100
Unemployment as % of civilian labour force	1.8	1.5	1.7	1.6	1.4	1.3	1.4	1.6	2.1	2.7	3.0
C. Other indicator											
Current balance (million US$)	1 510	969	817	1 223	1 602	1 757	1 476	1 947	1 762	2 039	. .

1. Throughout this *Survey*, GDP is on an SNA basis.
2. At constant 1985 prices.

Source: Statec, preliminary constant price data, *Rapport annuel 1995* and *Bulletin* No. 8-92; and OECD Secretariat estimates.

Table B. **Luxembourg – Gross domestic product**

Billion francs

	1986	1987	1988	1989	1990	1991	1992	1993	1994[1]	1995[1]
	Current prices									
Private consumption	157.5	167.2	179.7	195.5	214.6	234.5	240.2	254.4	266.6	274.8
Public consumption	32.2	35.7	37.5	40.9	46.4	49.5	53.2	57.5	60.9	67.2
Gross fixed capital formation	49.2	58.1	69.9	75.6	83.2	96.6	92.4	105.2	99.6	108.0
Change in stocks	–2.7	–7.0	–7.8	–0.2	–3.7	–1.3	–2.0	–8.3	4.5	3.9
Total domestic demand	236.2	253.9	279.3	311.8	340.5	379.3	383.8	408.8	431.6	453.9
Exports of goods and services	247.2	249.9	285.5	326.6	338.4	360.4	383.7	413.3	457.2	469.2
less: Imports of goods and services	232.7	244.8	277.0	311.3	333.2	367.3	361.8	377.8	401.2	411.8
Gross domestic product at market prices	250.7	259.0	287.9	327.2	345.7	372.4	405.7	444.3	487.7	511.2
	1985 prices									
Private consumption	156.7	163.9	171.5	180.2	190.4	202.5	200.6	204.0	208.9	. .
Public consumption	31.0	32.5	34.1	35.4	36.5	37.9	38.5	39.9	40.7	. .
Gross fixed capital formation	47.6	56.1	64.6	69.1	71.0	93.4	85.0	109.1	92.9	. .
Change in stocks	–3.9	–8.7	–10.1	–2.7	–6.8	–17.6	–12.5	–12.3	–3.1	. .
Total domestic demand	231.4	243.8	260.0	282.0	291.2	316.2	311.6	340.7	339.5	. .
Exports of goods and services	251.0	262.0	292.8	316.4	327.1	348.8	365.5	375.6	392.0	. .
less: Imports of goods and services	238.4	256.1	277.2	295.5	308.9	336.7	334.0	343.4	343.0	. .
Gross domestic product at market prices	244.0	249.7	275.7	302.8	309.3	328.4	343.0	372.9	388.6	. .

1. Estimates.

Source: Statec, *Note de conjoncture* No. 1-96 and preliminary data.

Table C. **Luxembourg – Labour market**

Thousands

	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995 [1]
Labour force [2] (A + B)	166.5	171.3	176.1	182.0	189.2	197.1	202.4	206.7	212.5	218.2
A. Unemployed	2.3	2.7	2.5	2.3	2.1	2.3	2.7	3.5	4.6	5.1
B. Total employment	164.2	168.6	173.6	179.7	187.1	194.8	199.7	203.2	207.9	213.1
Employees	146.3	151.0	156.3	162.7	170.4	178.4	183.4	187.0	191.8	197.1
Agriculture	1.3	1.3	1.3	1.4	1.5	1.5	1.5	1.5	1.6	1.6
Industry	38.7	38.4	37.1	37.7	37.5	37.8	37.3	35.7	34.7	34.2
of which: Iron and steel	15.4	14.5	13.5	13.1	12.3	11.6	10.9	10.0	9.4	8.9
Construction	13.5	14.6	15.9	16.8	18.3	20.2	21.6	22.1	21.9	22.1
Market services	68.2	71.5	76.0	80.5	85.9	90.8	94.1	97.8	102.7	107.4
Non-market services	24.6	25.2	26.0	26.4	27.1	28.1	28.9	29.8	30.8	31.8
Self-employed and family helpers	17.9	17.6	17.3	17.0	16.7	16.4	16.3	16.2	16.1	16.0
Unemployment rate (per cent)	1.5	1.7	1.6	1.4	1.3	1.4	1.6	2.1	2.7	3.0
Participation rate (per cent)	60.3	61.0	61.2	61.8	62.1	62.1	61.7	61.4	61.7	..
Job vacancies [3]										
Unfilled vacancies (monthly average)	234.6	244.9	215.9	200.7	177.4	153.2	158.1	124.7	270.5	437.5

1. Estimates.
2. Including border workers, net.
3. In units (not thousands).

Source: Statec, *Note de conjoncture* No.1-96; OECD, *Main economic indicators* and OECD Secretariat.

Table D. **Luxembourg – Structure of output and performance indicators**

1. Structure of output and economic performance (constant prices)

	1985	1990	1992	1993	1994	1995	1985	1990	1992	1993	1994	1995
	Share of GDP						Share of total employment					
Structure of output												
Agriculture, forestry and fishing	2.4	1.7	1.6	1.4	1.3	..	4.4	3.3	2.9	2.8	2.8	..
Energy and water	2.1	1.9	1.9	1.8	1.7	..	0.8	0.7	0.7	0.7	0.7	..
Mining, quarrying and manufacturing	26.1	23.4	23.8	20.9	19.4	..	24.0	19.6	18.2	17.0	16.5	..
of which: ores and metals	12.2	9.1	9.5	7.7	6.1	..	9.6	6.4	5.3	4.7	4.5	..
Construction	5.1	5.4	5.2	5.0	5.0	..	8.7	10.1	10.9	11.5	11.5	..
Market services [1]	64.9	74.2	76.5	79.6	80.6	..	47.8	52.6	54.7	54.8	55.2	..
of which: financial institutions and insurance companies	24.4	33.2	34.5	36.5	37.3	..	6.8	9.0	9.0	9.2	9.1	..
Non-market services [2]	11.9	10.2	9.7	9.4	9.2	..	14.3	13.9	14.3	14.3	14.2	..
Other [3]	–12.5	–16.9	–18.6	–18.1	–17.2	..	..	..	..	..	..	..
	Share of total investment						Productivity growth [4]					
Economic performance												
Agriculture, forestry and fishing	..	..	..	..	..	..	2.8	0.9	11.0	–1.0	–1.3	..
Energy and water	..	..	..	..	..	..	3.4	1.6	–3.1	10.7	2.1	..
Mining, quarrying and manufacturing	..	..	..	..	..	..	7.0	–2.1	10.3	0.4	–1.4	..
of which: ores and metals	..	..	..	..	..	..	6.4	0.5	18.8	–1.5	–15.3	..
Construction	..	..	..	..	..	..	1.2	0.3	–5.7	–2.1	1.1	..
Market services [1]	..	..	..	..	..	..	5.0	–0.6	4.0	10.8	4.0	..
of which: financial institutions and insurance companies	..	..	..	..	..	..	4.5	–0.2	12.2	10.7	6.4	..
Non-market services [2]	..	..	..	..	..	..	0.7	0.4	–3.2	3.7	1.6	..

1. Wholesale and retail trade; restaurants and hotels; transport, storage and communications; finance, insurance, real estate and business services.
2. Community, social and personal services.
3. Consumption of imputed banking services.
4. Average for 1985-89 instead of 1985.

Source: Statec, preliminary constant price data.

Table D. **Luxembourg – Structure of output and performance indicators** *(cont.)*

2. Other indicators

	1980	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Breakdown of employed workforce by size of establishment:[1]												
1 to 19 employees	13.6	14.4	13.7	13.8	13.9	13.9	13.7	15.0	14.9	16.1	16.8	16.7
20 to 49 employees	9.2	10.2	10.3	11.0	11.5	11.9	12.7	13.0	14.4	14.2	14.6	15.3
50 to 99 employees	8.5	9.2	9.6	9.6	10.4	11.7	11.2	10.1	10.4	12.0	11.4	10.5
100 to 249 employees	14.3	16.0	16.9	15.4	14.9	14.8	15.0	17.1	15.8	17.4	17.2	18.3
250 to 499 employees	12.0	11.9	13.1	8.7	9.6	8.7	10.1	10.8	12.5	11.0	10.9	12.2
500 employees and over	42.3	38.3	36.4	41.5	39.7	39.0	37.3	34.0	32.0	29.3	29.0	27.0
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Workforce (thousands)	57.4	52.0	53.3	54.2	55.0	56.0	56.5	59.8	60.3	59.8	58.4	59.2

1. Only industry and construction.

Source: Statec.

Table E. **Luxembourg – Public sector**

	1985	1990	1991	1992	1993	1994	1995
Budgetary indicators:							
general government accounts							
(as a per cent of GDP)							
Primary receipts [1]	49.6	..	..	..	..	..	..
Primary expenditure [1]	46.0	..	..	..	..	..	..
Primary balance [1]	3.6	..	..	..	..	..	..
Net interest payments	–1.2	..	..	..	..	..	..
General government budget balance	4.8	5.0	1.9	0.8	1.7	2.6	1.5
Structure of expenditure and taxes							
(as a per cent of GDP)							
General government current expenditure	47.1	..	..	..	..	..	..
of which:							
Subsidies	3.5	3.2	3.3	3.1	3.0	3.0	2.2
Social security benefits	22.0	21.8	22.8	23.0	23.6	..	..
Investment	4.4	4.6	4.9	5.5	5.5	4.4	4.6
Tax receipts	46.7	43.6	42.9	42.5	44.3	45.0	..
Personal income tax	12.0	10.3	9.5	9.2	9.2	9.7	..
Corporate tax	8.3	6.9	6.5	5.4	7.2	7.6	..
Social security contributions	12.3	11.9	12.1	12.6	12.4	12.0	..
Consumption taxes	11.4	10.8	11.5	12.1	12.1	12.3	..
of which: Value added tax	5.8	5.9	6.2	6.1	6.3	6.1	..
Other indicator							
Income tax as a per cent of total tax	43.3	39.4	37.3	34.2	37.0	38.4	..

	Prior to		After
Tax rates (per cent)			
Personal income tax rates			
Top rate	56	6th December 1990	50
Lowest rate	10	6th December 1990	10
Number of brackets	25	6th December 1990	18
Corporate tax rates	34	6th December 1990	33
VAT rates			
Lower reduced rate		1st January 1992	3
Reduced rate		1st January 1992	6
''Parking'' rate		1st January 1992	12
Standard rate	12	1st January 1992	15

1. Excluding interest charges.

Source: Statec, *Rapport annuel 1995* and *Annuaire statistique 1995;* Ministry of Finance; OECD, *Revenue Statistics 1965-1995* and *National Accounts*.

Table F. **Luxembourg – Financial markets**

	1980	1985	1990	1991	1992	1993	1994	1995
Sector size								
Sector employment[1]/total employment	4.8	6.3	8.7	8.8	8.8	9.1	9.5	9.4
Financial assets/GDP[2]	26.7	33.7	36.3	34.4	36.2	36.4	36.8	35.8
Stock-market capitalisation/GDP	30.9	59.5	99.5	99.0	94.4	177.1	220.3	216.3
Density of banking network[3]	30.5	32.2	46.7	48.6	54.6	55.2	55.4	53.7
Structure of financial flows								
Share of credits granted to non-financial sector in total banking assets	35.1	33.4	24.0	24.4	24.7	23.8	18.4	18.1
Internationalisation of markets								
Foreign business of the banking sector:[4]								
Assets	84.2	86.1	88.1	87.9	86.6	81.5	81.9	80.8
Liabilities	78.4	77.6	76.5	75.9	73.7	73.4	73.0	72.4
International banking network:								
Foreign banks in Luxembourg[5]	100	107	158	169	205	210	214	212
Luxembourg bank branches abroad[6]	74	70	55	6	6	7	7	11
Efficiency of markets								
Interest margins[7]	0.8	1.2	0.8	0.8	0.8	0.7	0.7	0.7
Bank productivity[8]	40.5	27.9	37.3	40.5	39.4	38.0	44.8	46.3
Cost of bank intermediation[9]	0.9	1.4	1.2	1.2	1.1	1.2	1.1	1.1

1. Credit institutions until 1990, credit institutions plus other intermediaries from 1991.
2. Ratio of banks' balance sheet total to GDP.
3. Number of saving and banking institutions per 100 000 population.
4. As a percentage of deposit banks' balance sheets.
5. Number of foreign saving and banking institutions.
6. Number of regional offices, agencies and branches abroad.
7. Interest margins divided by total assets.
8. Operating expenses as a per cent of gross earnings.
9. Gross earnings as a per cent of total assets.

Source: OECD, *Bank Profitability 1985-1994*; Statec, *Rapport annuel 1995* and data provided by the Institut monétaire Luxembourgeois.

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Greece
Population											
Total	Thousands	1994	17 840	8 031	10 124	29 251	5 206	5 088	57 960	81 407	10 430
Inhabitants per sq. km	Number	1994	2	96	332	3	121	15	106	228	79
Net average annual increase over previous 10 years	%	1994	1.4	0.6	0.3	1.6	0.2	0.4	0.5	2.9	0.5
Employment											
Total civilian employment (TCE)[2]	Thousands	1994	7 680 (93)	3 737	3 724 (92)	13 292	2 508	2 015	21 781 (93)	35 894	3 790
of which: Agriculture	% of TCE	1994	5.3 (93)	7.2	2.6 (92)	4.1	5.1	8.3	5.1 (93)	3.3	20.8
Industry	% of TCE	1994	23.7 (93)	33.2	27.7 (92)	22.6	26.8	26.8	27.7 (93)	37.6	23.6
Services	% of TCE	1994	71 (93)	59.6	69.7 (92)	73.3	68.1	64.9	67.2 (93)	59.1	55.5
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1994	331.6	198.1	227.9	544	146.7	97.2	1 328.5	1 832.3	73.1 (93)
Per capita	US$	1994	18 588	24 670	22 515	18 598	28 181	19 106	22 944	27 826	7 051 (93)
At current prices using current PPP's[3]	Bill. US$	1994	327.9	162.3	204.2	596.7	107	82.5	1 111.8	1 601.7	118
Per capita	US$	1994	18 382	20 210	20 166	20 401	20 546	16 208	19 201	24 325	1 450
Average annual volume growth over previous 5 years	%	1994	2.2	2.5	1.6	1.1	1.9	–1.6	1.1	2.6	1.4 (93)
Gross fixed capital formation (GFCF)	% of GDP	1994	21.4	24.8	17.4	18.7	14.8	14.3	18.1	18.5	17.4 (93)
of which: Machinery and equipment	% of GDP	1993	9.8	8.7	7.8	6.2	7.2	5.90	8.10	7.5	7.8
Residential construction	% of GDP	1993	5.2	6.3	4.6	6.1	2.9	3.7	4.9	6.5	3.6
Average annual volume growth over previous 5 years	%	1994	0.8	3.7	0.4	–0.1	–2.8	–12.9	–1	0.8	2.7 (93)
Gross saving ratio[4]	% of GDP	1994	16.8	25.3	22	16	17	16.6	19	21	15.5 (93)
General government											
Current expenditure on goods and services	% of GDP	1994	17.5	18.8	15	20.2	25.3	22.4	19.6	17.7	19.1 (93)
Current disbursements[5]	% of GDP	1993	36.9	48.4	55.3	49	61.1	58.9	51.5	45.6	51.2
Current receipts	% of GDP	1993	33.5	48.6	50.1	43	58.3	52.5	46.8	45.7	40.2
Net official development assistance	% of GNP	1993	0.34	0.31	0.41	0.46	1.03	0.76	0.66	0.44	..
Indicators of living standards											
Private consumption per capita using current PPP's[3]	US$	1993	10 803	10 546	12 090	11 863	10 042	8 814	11 395	10 733	6 367
Passenger cars, per 1 000 inhabitants	Number	1990	430	382	387	469	311	386	413	480[8]	169
Telephones, per 1 000 inhabitants	Number	1991	464	432	410	586	577	544	511	420[8]	413
Television sets, per 1 000 inhabitants	Number	1991	480	478	451	639	536	501	407	556[8]	197
Doctors, per 1 000 inhabitants	Number	1993	2.2 (91)	2.3	3.7	2.2	2.8 (92)	2.6 (92)	2.8	3.2 (92)	3.8 (92)
Infant mortality per 1 000 live births	Number		6.1	6.5	8	6.8	5.4	4.4	6.5	5.8	8.5
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1994	3	5.5	3.7	3.3	3.5	4.8	3.5	5.2	14.6
Consumer prices	%	1994	3	3.4	2.8	2.8	2.1	3.3	2.5	3.3	16.2
Foreign trade											
Exports of goods, fob*	Mill. US$	1994	47 363	44 881	137 259[7]	165 358	41 850	29 514	235 337	422 243	8 958
As % of GDP	%	1994	14.3	22.7	60.2	30.4	28.5	30.4	17.7	23	11.5 (93)
Average annual increase over previous 5 years	%	1994	5	6.7	6.5	7.1	8.3	4.9	5.6	4.4	3.4
Imports of goods, cif*	Mill. US$	1994	49 731	55 071	126 006[7]	148 297	35 932	23 091	220 508	376 566	21 111
As % of GDP	%	1994	15	27.8	55.3	27.3	24.5	23.8	16.6	20.6	30.1 (93)
Average annual increase over previous 5 years	%	1994	4	7.2	5	5.4	6.1	–1.3	3.5	6.9	5.4
Total official reserves[6]	Mill. SDRs	1994	7 730	11 523	9 505[7]	8 416	6 203	7 304	17 986	52 994	9 924
As ratio of average monthly imports of goods	Ratio	1994	1.9	2.5	0.9	0.7	2.1	3.8	1	1.7	5.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Iceland	Ireland	Italy	Japan	Luxembourg	Mexico	Netherlands	New Zealand
Population										
Total	Thousands	1994	267	3 571	57 190	124 960	398	93 010	15 382	3 526
Inhabitants per sq. km	Number	1994	3	51	190	331	153	47	377	13
Net average annual increase over previous 10 years	%	1994	1.1	0.1	0	0.4	0.8	2	0.6	0.8
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	138	1 168 (93)	20 152 (93)	64 530	162 (91)	32 439	6 631	1 560
of which: Agriculture	% of TCE	1994	9.4	12.7 (93)	7.5 (93)	5.8	3.7 (91)	25.8	4	10.4
Industry	% of TCE	1994	26.1	27.7 (93)	33 (93)	34	31.5 (91)	22.2	23	25
Services	% of TCE	1994	65.2	59.7 (93)	59.6 (93)	60.2	64.8 (91)	52.1	73	64.6
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	6.2	52	1 017.8	4 590	10.6 (92)	371.2	334.3	51.2
Per capita	US$	1994	23 199	14 550	17 796	36 732	27 073 (92)	3 991	21 733	14 513
At current prices using current PPP's[3]	Bill. US$	1994	5.1	54.3	1 068.4	2 593.7	11.7	673.3	285.9	57.3
Per capita	US$	1994	19 271	15 212	18 681	20 756	29 454	7 239	18 589	16 248
Average annual volume growth over previous 5 years	%	1994	0.6	4.7	1	2.1	4.1 (92)	3	2.3	2.5
Gross fixed capital formation (GFCF)	% of GDP	1994	15.2	15.1	16.4	28.6	20.4 (93)	20.7	19.3	20
of which: Machinery and equipment	% of GDP	1993	3.9	6.3	7.4	11.5	..	9.4	8.6	9.3
Residential construction	% of GDP	1993	4.4	4.1	5.3	5.4	..	4.9	5.1	4.9
Average annual volume growth over previous 5 years	%	1994	–4	1	–2.3	1.4	6.5 (92)	7.7	0.4	5.8
Gross saving ratio[4]	% of GDP	1994	16.9	19.5	18.8	31.2	60.2 (92)	15.1	24.4	20.7
General government										
Current expenditure on goods and services	% of GDP	1994	20.6	16	17.1	9.8	17.1 (92)	11.8[9]	14.2	14.7
Current disbursements[5]	% of GDP	1993	34.9	..	53.2	26.9	..	..	55.4	..
Current receipts	% of GDP	1993	35.9	..	47.1	32.9	..	..	54.5	..
Net official development assistance	% of GNP	1993	..	0.15	0.42	0.27	0.34 (92)	..	0.88	0.22
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	11 546	7 750	11 029	11 791	15 545	4 853	10 726	9 266
Passenger cars, per 1 000 inhabitants	Number	1990	464	228	478	282	470	85	356	440
Telephones, per 1 000 inhabitants	Number	1991	527	300	400	454	511	70	477	436
Television sets, per 1 000 inhabitants	Number	1991	319	276	421	613	267	148	485	443
Doctors, per 1 000 inhabitants	Number	1993	3	1.7 (92)	1.7 (91)	1.7 (92)	2.1 (92)	1	2.5 (90)	2
Infant mortality per 1 000 live births	Number		4.8	5.9	7.3	4.5 (92)	8.5 (92)	18	6.3	7.3
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	..	4.6	5.9	2.4	..	5.3	3.2	2.1
Consumer prices	%	1994	6.3	2.7	5.2	2	3.1	16.1	2.8	2.5
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	1 628	34 125	189 802	396 149	..	60 882	155 084	12 169
As % of GDP	%	1994	26.3	65.7	18.6	8.6	..	16.4	46.4	23.8
Average annual increase over previous 5 years	%	1994	2.7	10.5	6.2	7.6	..	21.7	7.6	6.5
Imports of goods, cif*	Mill. US$	1994	1 464	25 812	167 690	274 916	..	79 346	139 800	11 859
As % of GDP	%	1994	23.6	49.7	16.5	6	..	21.4	41.8	23.2
Average annual increase over previous 5 years	%	1994	0.7	8.1	1.9	5.5	..	25.5	6.1	6.1
Total official reserves[6]	Mill. SDRs	1994	201	4 189	22 102	86 214	..	4 301	23 655	2 540
As ratio of average monthly imports of goods	Ratio	1994	1.6	1.9	1.6	3.8	..	0.7	2	2.6

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population										
Total	Thousands	1994	4 337	9 900	39 150	8 781	6 994	60 573	58 375	260 651
Inhabitants per sq. km	Number	1994	13	107	78	20	169	78	238	28
Net average annual increase over previous 10 years	%	1994	0.5	–0.1	0.2	0.5	0.7	2.1	0.3	1
Employment										
Total civilian employment (TCE)[2]	Thousands	1994	1970 (93)	4 372	11 760	3 926	3 772	19 664	25 044 (93)	123 060
of which: Agriculture	% of TCE	1994	5.6 (93)	11.5	9.8	3.4	4	44.8	2.2 (93)	2.9
Industry	% of TCE	1994	23.1 (93)	32.8	30.1	25	28.8	22.2	26.2 (93)	24
Services	% of TCE	1994	71.3 (93)	55.7	60.2	71.6	67.2	33	71.6 (93)	73.1
Gross domestic product (GDP)										
At current prices and current exchange rates	Bill. US$	1994	103.4 (93)	87	482.4	196.6	257.3	130.7	1 019.5	6 649.8
Per capita	US$	1994	23 984 (93)	8 792	12 321	22 389	36 790	2 157	17 468	25 512
At current prices using current PPP's[3]	Bill. US$	1994	95.3	122	531.7	153	167.4	319.3	1 030.2	6 649.8
Per capita	US$	1994	21 968	12 335	13 581	17 422	23 942	5 271	17 650	25 512
Average annual volume growth over previous 5 years	%	1994	2.1 (93)	1.4	1.5	–0.3	0.5	3.6	0.8	2.1
Gross fixed capital formation (GFCF)	% of GDP	1994	22 (93)	25.7	19.8	13.7	22.8	24.5	15	17.2
of which: Machinery and equipment	% of GDP	1993	..	..	5.7	5.7	7.5	10.3	7.3	7.7
Residential construction	% of GDP	1993	..	..	4.1	4	14.9[10]	9.1	3.1	4
Average annual volume growth over previous 5 years	%	1994	–3.93	2.7	–1.2	–7.6	–0.4	5.1	–2.1	4.6
Gross saving ratio[4]	% of GDP	1994	21.9 (93)	24.2	18.8	13.7	29.3	22.5	13.5	16.2
General government										
Current expenditure on goods and services	% of GDP	1994	22.1 (93)	17.2	16.9	27.3	14.1	11.7	21.6	16.4
Current disbursements[5]	% of GDP	1993	..	..	43.7	67.3	36.7	..	42.7	35.8
Current disbursements[5]	% of GDP	1993	..	..	40.1	59	36	..	36.8	31.7
Current receipts	% of GDP	1993	1.23	0.36	0.32	1.33	0.49	..	0.34	0.19
Net official development assistance	% of GNP	1993	9 826	7 780	8 412	9 240	13 730	3 617	10 942	16 444
Indicators of living standards										
Private consumption per capita using current PPP's[3]	US$	1993	378	260	307	418	441	29	361	568
Passenger cars, per 1 000 inhabitants	Number	1990	515	273	340	687	603	143	445	553
Telephones, per 1 000 inhabitants	Number	1991	423	187	400	468	406	175	434	814
Television sets, per 1 000 inhabitants	Number	1991	3.2 (92)	2.9	4.1	3	3	0.9	1.5 (92)	2.3 (92)
Doctors, per 1 000 inhabitants	Number	1993	5	8.7	7.6	4.8	5.6	52.6	6.6	8.5 (92)
Infant mortality per 1 000 live births	Number		4	..	7.2	5.4	..	..	6.7	2.8
Wages and prices (average annual increase over previous 5 years)										
Wages (earnings or rates according to availability)	%	1994	2.7	9	5.6	5.7	3.9	73	4.6	3.6
Consumer prices	%	1994	34 645	17 072	73 129	61 122	70 467	18 456	205 170	512 627
Foreign trade										
Exports of goods, fob*	Mill. US$	1994	30.9 (93)	19.6	15.2	31.1	27.4	14.1	20.1	7.7
As % of GDP	%	1994	5	6.1	10.5	3.4	6.4	9.5	6.1	7.1
Average annual increase over previous 5 years	%	1994	27 345	25 967	92 182	51 730	68 126	22 976	227 026	663 256
Imports of goods, cif*	Mill. US$	1994	23.3 (93)	29.9	19.1	26.3	26.5	17.6	22.3	10
As % of GDP	%	1994	2.9	6.6	5.2	1	3.2	37.9	2.8	7
Average annual increase over previous 5 years	%	1994	13 033	10 627	28 475	15 929	23 790	4 911	28 094	43 350
Total official reserves[6]	Mill. SDRs	1994	5.7	4.9	3.7	3.7	4.2	2.6	1.5	0.8

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. Gold included in reserves is valued at 35 SDRs per ounce. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Refers to the public sector including public enterprises.
10. Including non-residential construction.

Sources: Population and Employment: OECD, *Labour Force Statistics*. GDP, GFCF, and General Government: OECD, *National Accounts*, Vol. I and *OECD Economic Outlook*, Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators*. Foreign trade: OECD, *Monthly Foreign Trade Statistics*, series A. Total official reserves: IMF, *International Financial Statistics*.

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